



Local Holomorphy and Global Algebraicity: The Road to GAGA

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Introduction

A Riemann surface is a Hausdorff topological space locally homeomorphic to the complex plane, in which the transition maps between overlapping charts are biholomorphic. This apparently simple definition proves remarkably fertile: it provides a natural setting where the tools of complex analysis, differential geometry, and algebraic geometry all interact, and it offers the right intuition for the later study of higher-dimensional complex manifolds. The reference [GH94] treats the general case, but can feel quite overwhelming at first encounter. This report aims to serve as a more accessible entry point into this circle of ideas, focusing on the case of Riemann surfaces.

The central goal of this text is to present a version of Serre's GAGA theorem — an acronym for Géométrie Algébrique et Géométrie Analytique — in the special case of compact Riemann surfaces. This theorem, proved by Serre in 1956, establishes a deep equivalence between two seemingly different ways of studying the same geometric object. One may view a compact complex manifold through the lens of complex analysis, examining the holomorphic functions and differential forms defined locally on it, or consider the zero locus of homogeneous polynomials in some projective space and work algebraically, using rational maps and polynomial equations. The GAGA theorem asserts that, for smooth projective varieties, these two perspectives are in fact equivalent.

The proof relies on two major results. The first is the Riemann–Roch theorem, which links the dimension of spaces of meromorphic functions with prescribed poles to the topology of the surface, as encoded by its genus. It is itself a consequence of a study of sheaves and their cohomology, and of Serre duality. The second is Chow's theorem, which guarantees that any closed analytic subset of projective space is actually algebraic, that is, cut out by finitely many polynomial equations. Together, these results show that compact Riemann surfaces and smooth projective algebraic curves are one and the same class of objects.

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I. Prolegomena

Throughout this section, the basic definitions and properties of Riemann surfaces and maps between them will be developed.

I.1. Riemann Surfaces

Formally, a Riemann surface looks locally like the complex plane; the purpose of this subsection is to make this statement precise.

I.1.1. Definition A *local chart* on a topological space X consists of an open set U and a homeomorphism $\varphi : U \to V$, where V is an open subset of $\mathbb{C}$. It is said to be *centered at* p if $p \in U$ and $\varphi(p) = 0$.

This allows us to define complex coordinates on U, but our goal is to describe X itself. For this, we introduce the following definition.

I.1.2. Definition An *atlas* on a topological space X is a family of homeomorphisms from open sets of X to open sets of $\mathbb{C}$

$$\{\varphi_\alpha : U_\alpha \to V_\alpha\}$$

such that

$$X = \bigcup_\alpha U_\alpha.$$

We additionally require the coordinate change

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

to be biholomorphic. The charts $\varphi_\alpha : U_\alpha \to V_\alpha$ and $\varphi_\beta : U_\beta \to V_\beta$ are then said to be *compatible*.

Thus X is well described locally, but another choice of atlas could do so equally well. Two atlases are said to be *equivalent* if their union is again an atlas. This is an equivalence relation, and an equivalence class is called a *complex structure*.

I.1.3. Definition A *Riemann surface* is a connected, Hausdorff, second-countable topological space endowed with a complex structure.

I.1.4. Example The Riemann sphere $\mathbb{C}_\infty := \mathbb{C} \cup \{\infty\}$ is the Hausdorff compactification of the complex plane. It is a Riemann surface with the local charts : z on $\{z \neq \infty\}$ and $w = 1/z$ on $\{z \neq 0\}$.

I.1.5. Remark The Riemann sphere $\mathbb{C}_\infty$ admits a natural algebraic description: it is the projective line $\mathbb{P}^1 = (\mathbb{C}^2 \setminus \{0\})/\sim$, where $(Z_0 : Z_1) \sim (\lambda Z_0 : \lambda Z_1)$ for $\lambda \in \mathbb{C}^\times$. The identification sends $z \in \mathbb{C}$ to $(z : 1)$ and ∞ to $(1 : 0)$. This point of view, which places $\mathbb{C}_\infty$ as the simplest instance of a projective space, will be central later on.

I.2. Maps on Riemann Surfaces

A useful idea to study the topology of a space is to look for maps from it. On Riemann surfaces, since they are locally homeomorphic to the complex plane, one is interested in the study of holomorphic maps.

I.2.1. Definition Let X be a Riemann surface, with a given complex structure $\{\varphi_\alpha\}$, $f : X \to \mathbb{C}$ is said to be *holomorphic* (resp. *meromorphic*) if every $f \circ \varphi_\alpha^{-1}$ is holomorphic (resp. meromorphic).

We first need to check that it is well defined : independent of the choice of atlas in the complex structure. Given two choices of local charts $\varphi_\alpha, \varphi_\beta$ such that $U_\alpha \cap U_\beta \neq \varnothing$, one can see that

$$f \circ \varphi_\alpha^{-1} = f \circ \varphi_\beta^{-1} \circ \underbrace{\left(\varphi_\beta \circ \varphi_\alpha^{-1}\right)}_{\text{holomorphic}},$$

and thus if $f \circ \varphi_\alpha^{-1}$ is holomorphic (resp meromorphic) then so is $f \circ \varphi_\beta^{-1}$ and conversely (since $\varphi_\beta \circ \varphi_\alpha^{-1}$ is biholomorphic).

Since those functions are locally holomorphic functions from the complex plane, most of the properties of complex holomorphic functions adapt.

I.2.2. Definition [Order of a meromorphic function] Let f be a meromorphic function on a Riemann surface X, and let $p \in X$. Choose a local coordinate z centered at p. Then f admits a Laurent expansion

$$f(z) = \sum_{k=m}^{\infty} a_k z^k, \qquad a_m \neq 0.$$

The integer $m \in \mathbb{Z}$ is called the *order* of f at p, and is denoted $\mathrm{ord}_p(f)$. If $f = 0$ then let $m = \infty$.

- If $m \geq 0$, then p is a zero of order m.
- If $m < 0$, then p is a pole of order $-m$.

The following propositions are only consequences of their equivalent on open sets of the complex plane.

I.2.3. Proposition [The Identity Theorem] *Let X be a Riemann surface, and f, g two holomorphic functions on X, if $\{f = g\}$ is not discrete, then $f = g$.*

Proof. Let $p \in X$ be an accumulation point of $\{f = g\}$, and $\phi : U \to V$ a local chart centered at p. Then $f\phi^{-1} = g\phi^{-1}$ on V and thus $f = g$ on U. Now, since X is connected, and the set $\{f = g\}$ is open, closed and non-empty, $f = g$ on X □

I.2.4. Proposition [Maximum modulus principle] *Let X be a Riemann surface and $f : X \to \mathbb{C}$ a holomorphic function. If $|f|$ attains a local maximum at some point $p \in X$, then f is constant.*

Proof. Since f is locally a holomorphic function on the complex plane, choose a chart $\phi : U \to V$ centered at p. Then $f \circ \phi^{-1}$ attains a local maximum of its modulus at $\phi(p)$, and by the classical maximum modulus principle, it is constant on V. Hence f is constant on U,and thus on X by I.2.3. □

I.2.5. Corollary [Liouville] If X is compact, every holomorphic function on X happens to be constant.

I.2.6. Proposition [Zeros of holomorphic functions are isolated] *Let X be a Riemann surface and $f : X \to \mathbb{C}$ a non-constant holomorphic function. Then the following set has no accumulation point : $\{p \in X : f(p) = 0\}$.*

I.2.7. Proposition [Local form of meromorphic functions] *Let X be a Riemann surface and $f : X \to \mathbb{C} \cup \{\infty\}$ a meromorphic function. Then for every $p \in X$, there exists a chart $\phi : U \to V$ centered at p such that*

$$f \circ \phi^{-1}(z) = \frac{g(z)}{h(z)},$$

where g and h are holomorphic on V and $h \not\equiv 0$.

Proof. By definition, f is holomorphic on $X \setminus P$, where P is a discrete set of poles. If $p \notin P$, the statement is trivial. If $p \in P$, choose a chart $\phi : U \to V$ centered at p. Then $f \circ \phi^{-1}$ is meromorphic on V, hence by the classical theory of meromorphic functions on the plane, it can be written as a quotient of two holomorphic functions g and h with $h \not\equiv 0$. This gives the desired expression. □

I.2.8. Proposition [Order of a zero or pole] *Let X be a Riemann surface and f a meromorphic function on X. For every $p \in X$, there exists a chart $\phi : U \to V$ centered at p such that*

$$f \circ \phi^{-1}(z) = (z - \phi(p))^k \, u(z),$$

where $k \in \mathbb{Z}$ and u is holomorphic and non-vanishing on V. The integer k is called the order *of f at p and is denoted* $\operatorname{ord}_p(f)$.

Proof. Choose a chart $\phi : U \to V$ centered at p. If f is holomorphic at p, then $f \circ \phi^{-1}$ has a Taylor expansion

$$f \circ \phi^{-1}(z) = a_k(z - \phi(p))^k + \cdots,$$

with $a_k \neq 0$ for some $k \geq 0$, giving the desired form. If f has a pole at p, then $(1/f) \circ \phi^{-1}$ is holomorphic and vanishes at $\phi(p)$, so the previous case applied to $1/f$ yields the result with $k < 0$. In both cases, the holomorphic function u is non-vanishing near $\phi(p)$. □

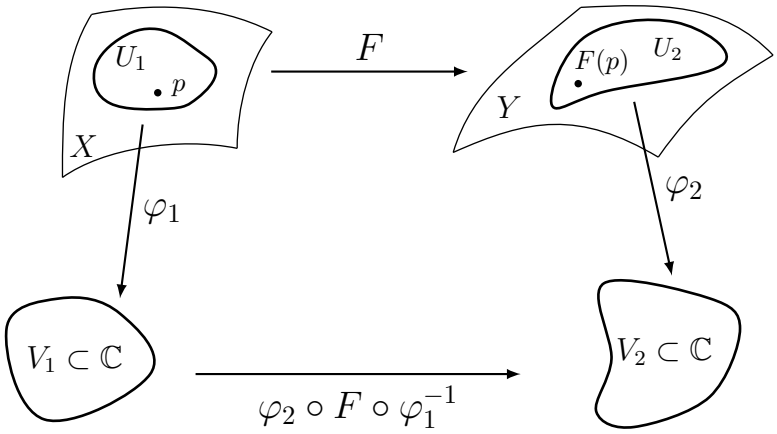
I.3. Holomorphic maps between Riemann surfaces

Once the notion of a Riemann surface has been introduced, the next natural step is to understand the appropriate class of mappings between them. Just as holomorphic functions are the fundamental objects of complex analysis on the plane, holomorphic maps play the same role in the category of Riemann surfaces. Since every Riemann surface is locally modeled on open subsets of $\mathbb{C}$, the correct notion of regularity for a map between two such surfaces should be expressed in terms of holomorphicity in local coordinates. The following definition formalizes this idea.

I.3.1. Definition Let X and Y be Riemann surfaces. A map $F : X \to Y$ is said to be *holomorphic at a point $p \in X$* if there exist charts $\phi_1 : U_1 \to V_1$ on X with $p \in U_1$ and $\phi_2 : U_2 \to V_2$ on Y with $F(p) \in U_2$ such that the composition

$$\phi_2 \circ F \circ \phi_1^{-1} : V_1 \longrightarrow V_2$$

is holomorphic at $\phi_1(p)$ in the usual complex-analytic sense. The following diagram sums up the situation.



If F is holomorphic at every point of an open set $W \subset X$, we say that F is *holomorphic on* W. In particular, F is a *holomorphic map* if it is holomorphic on all of X.

Again, most of the properties of complex holomorphic maps transfer to maps between Riemann surfaces. These are not proved here but they are basic applications of the properties on $\mathbb{C}$.

I.3.2. Proposition [Open Mapping Theorem] *Let $F : X \to Y$ be a nonconstant holomorphic map between Riemann surfaces. Then F is an open map.*

I.3.3. Proposition *Let $F : X \to Y$ be an injective holomorphic map between Riemann surfaces. Then F is an isomorphism between X and its image $F(X)$.*

I.3.4. Proposition [Identity Theorem] *Let $F, G : X \to Y$ be holomorphic maps between Riemann surfaces. If $F = G$ on a subset of X having a limit point in X, then $F = G$ on all of X.*

I.3.5. Proposition *Let X be a compact Riemann surface and $F : X \to Y$ a nonconstant holomorphic map. Then Y is compact and F is surjective.*

I.3.6. Proposition [Discreteness of Fibers] *Let $F : X \to Y$ be a nonconstant holomorphic map between Riemann surfaces. Then for every $y \in Y$, the fiber $F^{-1}(y)$ is a discrete subset of X. If moreover X and Y are compact, then $F^{-1}(y)$ is finite and nonempty.*

A meromorphic function on a Riemann surface may be interpreted as a holomorphic map to the Riemann sphere. Indeed, if f is meromorphic on X, it takes complex values away from its poles, while the natural value at a pole is ∞. This leads to the construction of a map $F : X \to \mathbb{C}_\infty$ defined by $F(x) = f(x)$ when x is not a pole of f, and $F(x) = \infty$ otherwise. One checks that this map is holomorphic in the sense of Riemann surfaces, and this viewpoint provides a precise correspondence between meromorphic functions and holomorphic maps to the sphere. We just proved the following proposition.

I.3.7. Proposition *There is a one-to-one correspondence between meromorphic functions on X and the set of holomorphic maps $F : X \to \mathbb{C}_\infty$ such that $F \not\equiv \infty$*

Holomorphic maps between Riemann surfaces exhibit a remarkably rigid local structure. Although global behavior can be subtle, the local picture is completely controlled: every nonconstant holomorphic map looks, in suitable coordinates, like a power map. This motivates the notion of multiplicity and provides the fundamental local invariant from which many global properties are derived.

I.3.8. Proposition [Local Normal Form] *Let $F : X \to Y$ be a holomorphic map defined at a point $p \in X$, and assume that F is not constant. Then there exists a unique*

integer $m \geq 1$ such that for every chart $\phi_2 : U_2 \to V_2$ on Y centered at $F(p)$, one can find a chart $\phi_1 : U_1 \to V_1$ on X centered at p satisfying

$$\phi_2\big(F(\phi_1^{-1}(z))\big) = z^m.$$

Proof. Fix a local chart $\phi_2 : U_2 \to V_2$ centered at $F(p)$. For any chart $\phi : U \to V$ centered at p, consider the Taylor expansion of $T := \phi_2 \circ F \circ \phi^{-1}$:

$$T(w) = \sum_{k=m}^{\infty} a_k w^k,$$

with $a_m \neq 0$. Thus we may write $T(w) = w^m S(w)$, where S is holomorphic and nonvanishing at 0. Since $S(0) \neq 0$, one can choose a holomorphic branch of the logarithm on a neighborhood of $S(0)$. Define

$$R(w) := \exp\left(\frac{1}{m} \log S(w)\right),$$

so that $R(w)^m = S(w)$ near 0, and R is holomorphic with $R(0) \neq 0$.

Define $\eta(w) = wR(w)$; this gives a new coordinate z on V, and by the inverse function theorem η is invertible near 0 since $\eta'(0) \neq 0$. Set $\phi_1 := \eta \circ \phi$. Then

$$\begin{aligned}
\phi_2\big(F(\phi_1^{-1}(z))\big) &= \phi_2 \circ F \circ \phi^{-1}(\eta^{-1}(z)) \\
&= T(w) \\
&= w^m S(w) \\
&= (wR(w))^m \\
&= z^m.
\end{aligned}$$

The uniqueness comes from the fact that coordinate changes preserve the order. □

The identification $\mathbb{C}_\infty \cong \mathbb{P}^1$ noted earlier is thus not merely a curiosity: it shows that holomorphic maps to the Riemann sphere are, in disguise, holomorphic maps into projective space. This motivates a systematic study of projective spaces as natural targets and ambient spaces for Riemann surfaces, which we now undertake.

I.4. Projective spaces

There are two classical ways to introduce the projective plane: an algebraic and a geometric one.

For the algebraic approach, consider the Fermat equation $x^n + y^n = z^n$. Looking for integer solutions is equivalent to looking for rational ones of $x^n + y^n = 1$. A key observation is that if (x, y, z) is a solution, then so is (mx, my, mz) for any non-zero integer m. These triples should therefore represent the same point: only their direction in $\mathbb{Q}^3$ matters. This motivates the definition of the projective plane $\mathbb{P}^2$ over the complex field $\mathbb{C}$ as

$$\mathbb{P}^2 = (\mathbb{C}^3 \setminus \{0\})/\sim,$$

where $[x : y : z] \sim [\lambda x : \lambda y : \lambda z]$ for all $\lambda \in \mathbb{C}^\times$. The reader may define $\mathbb{P}^n$ similarly.

For the geometric viewpoint, we want a space in which Bézout's theorem holds for lines. In the affine plane $\mathbb{C}^2$, the only obstruction comes from parallel lines, which never meet. Projective geometry fixes this by adding a point at infinity for each possible direction. This is the familiar phenomenon of the two sides of a straight road appearing to meet at a vanishing point. Thus, geometrically,

$$\mathbb{P}^2 = \mathbb{C}^2 \cup \{\text{directions of lines in } \mathbb{C}^2\}.$$

To make this precise, consider the affine chart $z = 1$. Any point $[x : y : z]$ with $z \neq 0$ can be uniquely written as

$$[x : y : z] = \big[x/z : y/z : 1\big],$$

which identifies the open subset $\{z \neq 0\}$ in $\mathbb{P}^2$ with the affine plane $\mathbb{C}^2$ via

$$(x, y) \longleftrightarrow [x : y : 1].$$

The complement of this chart is the line $\{z = 0\}$, called the *line at infinity*. It consists exactly of the directions of affine lines: each point $[x : y : 0]$ corresponds to a slope or direction in $\mathbb{C}^2$. Parallel affine lines meet at the point of $\{z = 0\}$ representing their common direction.

We therefore obtain the geometric decomposition

$$\mathbb{P}^2 = \{z \neq 0\} \cup \{z = 0\} = \mathbb{C}^2 \cup \mathbb{P}^1,$$

where $\mathbb{P}^1$ parameterizes the directions of lines in the affine plane. Thus $\mathbb{P}^1 \cong \mathbb{C}_\infty$

To make this clear, here is a recap array :

Algebraic definition		Geometric definition
$\mathbb{C}^3 \setminus \{0\}/\sim$	$\leftrightarrow$	$\mathbb{C}^2 \cup \mathbb{P}^1$
(a, b, c)	$\rightarrow$	$\begin{cases} (a/c, b/c) \in \mathbb{C}^2 & \text{if } c \neq 0 \\ (a, b) \in \mathbb{P}^1 & \text{if } c = 0 \end{cases}$
$(x, y, 1)$	$\leftarrow$	$(x, y) \in \mathbb{C}^2$
$(x, y, 0)$	$\leftarrow$	$(x, y) \in \mathbb{P}^1$

We now introduce the first fundamental notion: a **projective** (algebraic) curve. Let $f \in \mathbb{C}[x, y]$ be a polynomial of degree d, which we may write as

$$f(x, y) = \sum_{i+j \leq d} a_{i,j} x^i y^j.$$

Our goal is to extend the affine curve $\mathcal{C} = \{f = 0\}$ to a curve in $\mathbb{P}^2$. To do so, we homogenize f by defining

$$F(x, y, z) = \sum_{i+j \leq d} a_{i,j} x^i y^j z^{d-i-j}.$$

Then F is homogeneous of degree d, and satisfies

$$F(x, y, 1) = f(x, y).$$

For a point $(x : y : z) \in \mathbb{P}^2$, the equation $F(x, y, z) = 0$ is well defined: indeed, since $(x : y : z) \sim (tx : ty : tz)$ for any $t \in \mathbb{C}^\times$, we must have

$$F(tx, ty, tz) = 0 \quad \text{whenever} \quad F(x, y, z) = 0.$$

This holds precisely because F is homogeneous:

$$F(tx, ty, tz) = t^d F(x, y, z).$$

Thus $F = 0$ defines a projective curve in $\mathbb{P}^2$ which extends the affine curve $\mathcal{C}$. More precisely,

$$(x, y) \in \mathcal{C} \quad \Longleftrightarrow \quad (x : y : 1) \in \{(a : b : c) \in \mathbb{P}^2 \mid F(a, b, c) = 0\}.$$

The projective curve also contains additional points at infinity, namely those of the form $(x : y : 0)$ satisfying $F(x, y, 0) = 0$.

I.5. Integration on Riemann surfaces

One of the main tools in the theory of complex variables is contour integration. We want to transport this notion to Riemann surfaces, and the correct objects to integrate are differential forms.

I.5.1. Differential forms

I.5.1. Definition A holomorphic (resp. meromorphic, $\mathcal{C}^\infty$) 1-form on an open set $V \subset \mathbb{C}$ is an expression of the form

$$\omega = f(z)\, dz,$$

where f is holomorphic (resp. meromorphic, $\mathcal{C}^\infty$) and dz denotes the differential of the complex coordinate function z.

I.5.2. Definition Let $\omega_1(z) = f_1(z)\, dz$ be a 1-form on V_1 and $\omega_2(w) = f_2(w)\, dw$ a 1-form on V_2. If $T : V_2 \to V_1$ is holomorphic, we say that ω_1 *transforms to* ω_2 under T if

$$f_2(w) = f_1(T(w))\, T'(w).$$

Of course, the definition we want is on Riemann surfaces. To obtain it, we use local charts.

I.5.3. Definition A holomorphic (resp. meromorphic, $\mathcal{C}^\infty$) 1-form on a Riemann surface X is a collection of holomorphic (resp. meromorphic, $\mathcal{C}^\infty$) 1-forms $\{\omega_\alpha\}$ on the coordinate domains of the charts $\varphi_\alpha : U_\alpha \to V_\alpha$ such that whenever $U_\alpha \cap U_\beta \neq \varnothing$, the forms satisfy the compatibility condition

$$\omega_\alpha \text{ transforms to } \omega_\beta \text{ under } \varphi_\alpha \circ \varphi_\beta^{-1}.$$

I.5.4. Definition The order of a meromorphic 1-form ω on X at p is the order of f at 0 in the local coordinate : $\omega(z) = f(z)dz$

Defining a form on every chart of X can sometimes be complicated, thus, we use the identity theorem, and only define it on a local chart, leaving the existence problem to the reader.

I.5.5. Definition Let $F : X \to Y$ be a holomorphic map between Riemann surfaces and

$$\omega = f(z, \overline{z})dz + g(z, \overline{z})d\overline{z}$$

be a $\mathcal{C}^\infty$ 1-form on Y. The *pullback* of ω by F is the $\mathcal{C}^\infty$ 1-form on X defined by

$$F^*\omega = f(h(w), \overline{h(w)})h'(w)dw + g(h(w), \overline{h(w)})\overline{h'(w)}d\overline{w},$$

where $h = \varphi_2 \circ F \circ \varphi_1^{-1}$ is the local representative of F with respect to chosen charts.

I.5.6. Lemma *Let $F : X \to Y$ be a holomorphic map between Riemann surfaces, ω be a meromorphic 1-form on Y and $p \in X$. Then*

$$\operatorname{ord}_p F^*\omega = (1 + \operatorname{ord}_{F(p)} \omega) \cdot \operatorname{mult}_p F - 1$$

Proof. Let w denote the local coordinate at p and z the one at $F(p)$, chosen such near p, F has the form $z = w^n$ (with $n = \operatorname{mult}_p F$). At $F(p)$, ω can be written

$$\omega = (cz^k + \cdots)dz,$$

where $k = \operatorname{ord}_{F(p)} \omega$.

Using this we see that $F^*\omega$ has the local form at p

$$F^*\omega = \left(cnw^{nk+n-1} + \cdots\right) dw$$

and the formula is proven. □

I.5.2. C^∞-forms on Riemann Surfaces

Let X be a Riemann surface. Since X is locally modeled on open subsets of $\mathbb{C} \cong \mathbb{R}^2$, differential forms on X are defined by patching together the classical calculus on the plane.

Fix a local chart $\phi : U \to V \subset \mathbb{C}$, writing $z = x + iy$ for the standard complex coordinate on V. Introduce the *complex differentials*

$$dz := dx + i\,dy, \qquad d\bar{z} := dx - i\,dy,$$

so that $dx = \frac{1}{2}(dz + d\bar{z})$ and $dy = \frac{1}{2i}(dz - d\bar{z})$.

I.5.7. Definition [Smooth (p, q)-forms] Let X be a Riemann surface. In a local chart with coordinate z, a smooth differential form of type (p, q) is a finite sum of expressions of the form

$$f(z, \bar{z})\, dz^{\wedge p} \wedge d\bar{z}^{\wedge q},$$

where $f \in C^\infty(U, \mathbb{C})$ and $p, q \in \{0, 1\}$. Explicitly:

$$\begin{aligned}
(0,0)\text{-forms}: &\quad f,\\
(1,0)\text{-forms}: &\quad f\,dz,\\
(0,1)\text{-forms}: &\quad f\,d\bar{z},\\
(1,1)\text{-forms}: &\quad f\,dz \wedge d\bar{z}.
\end{aligned}$$

I.5.8. Remark Note that $p, q < 2$ because $dz \wedge dz = d\bar{z} \wedge d\bar{z} = 0$ and the wedge products are cyclic and anti-symmetric.

I.5.9. Remark The decomposition into types is intrinsic to the complex structure: under a holomorphic change of coordinates $w = \psi(z)$,

$$dw = \psi'(z)\, dz, \qquad d\bar{w} = \overline{\psi'(z)}\, d\bar{z},$$

so (p, q)-forms transform among themselves. A $(1, 0)$-form is holomorphic if its coefficient f is holomorphic.

Spaces of Smooth, Holomorphic and Meromorphic Forms

Fix an open set $U \subset X$. We introduce the following spaces:

$$\begin{aligned}
\mathcal{E}_X^{(k)}(U) &= \{\, C^\infty \text{ } k\text{-forms on } U \,\},\\
\mathcal{E}_X^{(p,q)}(U) &= \{\, C^\infty \text{ } (p,q)\text{-forms on } U \,\},\\
\mathcal{O}_X(U) &= \{\, \text{holomorphic functions } f : U \to \mathbb{C} \,\},\\
\Omega_X^1(U) &= \{\, \text{holomorphic } (1,0)\text{-forms on } U \,\},\\
\mathcal{M}_X(U) &= \{\, \text{meromorphic functions on } U \,\},\\
\mathcal{M}_X^{(1)}(U) &= \{\, \text{meromorphic 1-forms on } U \,\}.
\end{aligned}$$

All of these spaces are complex vector spaces. The subscript X will sometimes be forgotten when it is clear. The spaces $\mathcal{E}_X(U)$, $\mathcal{O}_X(U)$, and $\mathcal{M}_X(U)$ are $\mathbb{C}$-algebras. If U is connected, then $\mathcal{O}_X(U)$ is an integral domain and $\mathcal{M}_X(U)$ is a field. The spaces, $\Omega_X^1(U)$ and $\mathcal{M}_X^{(1)}(U)$ are modules over $\mathcal{O}_X(U)$, and if U is connected, $\mathcal{M}_X^{(1)}(U)$ is a vector space over the field $\mathcal{M}_X(U)$.

The Exterior Derivative and its Decomposition

I.5.10. Definition The *exterior derivative* $d : \mathcal{E}_X^{(k)}(X) \to \mathcal{E}_X^{(k+1)}(X)$ is the unique $\mathbb{C}$-linear operator satisfying:

(i) $df = \partial f/\partial x\, dx + \partial f/\partial y\, dy$ for $f \in C^\infty(X)$;

(ii) $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \mathcal{E}^{(k)}$, $\beta \in \mathcal{E}^{(\ell)}$;

(iii) $d^2 = 0$.

On a Riemann surface, d decomposes according to the bigrading. Define the *Dolbeault operators*

$$\partial := \frac{\partial}{\partial z}\, dz, \qquad \bar\partial := \frac{\partial}{\partial \bar z}\, d\bar z,$$

with

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \frac{\partial}{\partial \bar z} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

I.5.11. Proposition [Decomposition of d] *For any C^∞ form ω on a Riemann surface,*

$$d = \partial + \bar\partial.$$

More precisely:

$$\begin{aligned}
d &: \mathcal{E}^{(0,0)} \to \mathcal{E}^{(1,0)} \oplus \mathcal{E}^{(0,1)},\\
d &: \mathcal{E}^{(1,0)} \to \mathcal{E}^{(1,1)},\\
d &: \mathcal{E}^{(0,1)} \to \mathcal{E}^{(1,1)},\\
d &: \mathcal{E}^{(1,1)} \to 0.
\end{aligned}$$

Furthermore, $\partial^2 = \bar\partial^2 = 0$ and $\partial\bar\partial + \bar\partial\partial = 0$.

Proof. Simple computations using the definition. □

I.5.12. Proposition [Formal rules for ∂ and $\bar\partial$] *Let X be a Riemann surface. If α and β are smooth forms of pure type, then*

$$\bar\partial(\alpha\wedge\beta) = \bar\partial\alpha\wedge\beta + (-1)^{\deg\alpha}\alpha\wedge\bar\partial\beta.$$

In particular, if $f\in\mathcal{E}_X^{(0,0)}(U)$ and $\eta\in\mathcal{E}_X^{(1,0)}(U)$, then

$$\bar\partial(f\eta) = \bar\partial f\wedge\eta + f\,\bar\partial\eta.$$

Proof. The statement is local, so it suffices to work in a coordinate $z=x+iy$ on an open subset of $\mathbb{C}$. They then follow from the definitions and the identity $d=\partial+\bar\partial$. □

I.5.13. Remark [Cut-off computation] Let $\rho\in\mathcal{E}_X^{(0,0)}(X)$ be a smooth function and let ω be a meromorphic 1-form on an open set $U\subset X$. Assume that $\rho\omega$ extends to a smooth $(1,0)$-form on X. Then

$$\bar\partial(\rho\omega) = \bar\partial\rho\wedge\omega$$

on $U\setminus\operatorname{Supp}(\text{poles}(\omega))$. Indeed,

$$\bar\partial(\rho\omega) = \bar\partial\rho\wedge\omega + \rho\,\bar\partial\omega = \bar\partial\rho\wedge\omega,$$

since ω is holomorphic away from its poles.

Closed and Exact Forms

I.5.14. Definition A k-form ω is *closed* if $d\omega=0$, and *exact* if $\omega=d\alpha$ for some $(k-1)$-form α. Since $d^2=0$, every exact form is closed.

I.5.15. Proposition [Holomorphic forms are closed] *Every holomorphic 1-form on a Riemann surface is closed.*

Proof. Let $\omega=f\,dz$ be a holomorphic $(1,0)$-form, so f is holomorphic. Then

$$d\omega = \frac{\partial f}{\partial\bar z}\,d\bar z\wedge dz.$$

Since f is holomorphic, the Cauchy–Riemann equations give $\frac{\partial f}{\partial\bar z}=0$, hence $d\omega=0$. □

I.5.16. Proposition [Poincaré lemma, smooth version] *Let $U\subset X$ be an open disc (contractible open set). Then every closed k-form on U is exact.*

Proof. It suffices to treat $U=\mathbb{D}\subset\mathbb{C}$, the open unit disc, with coordinates (x,y).

Case $k=1$. Let $\omega=p\,dx+q\,dy$ be a closed 1-form on $\mathbb{D}$, so $\partial_y p=\partial_x q$. Define

$$f(x,y) := \int_0^1 \big[p(tx,ty)\,x + q(tx,ty)\,y\big]\,dt.$$

One verifies by direct differentiation under the integral sign that $\frac{\partial f}{\partial x}=p$ and $\frac{\partial f}{\partial y}=q$, using the closedness condition $\partial_y p=\partial_x q$. Hence $\omega=df$.

Case $k = 2$. Let $\omega = h\,dx \wedge dy$ be a 2-form on $\mathbb{D}$. Define

$$\alpha(x, y) := \left(\int_0^1 h(tx, ty)\, t\, dt\right)(-y\, dx + x\, dy).$$

A direct computation gives $d\alpha = \omega$: differentiating and applying the product rule, one obtains $h\,dx \wedge dy$ after using the chain rule for $h(tx, ty)$. Hence every 2-form is exact on $\mathbb{D}$. □

I.5.17. Proposition [Stokes' theorem on a Riemann surface] *Let X be a Riemann surface, let $U \subset X$ be a relatively compact domain with smooth boundary, and let α be a smooth 1-form on a neighborhood of $\overline{U}$. Then*

$$\int_U d\alpha = \int_{\partial U} \alpha.$$

I.5.18. Remark If $\eta \in \mathcal{E}_X^{(1,0)}(U)$, then $d\eta = \bar\partial\eta \in \mathcal{E}_X^{(1,1)}(U)$, since $\partial\eta = 0$ for bidegree reasons. Hence Stokes' theorem gives

$$\int_U \bar\partial\eta = \int_{\partial U} \eta.$$

This is the form of Stokes' theorem most often used in residue computations.

I.5.19. Proposition [Local solvability of $\bar\partial$] *Let $D \subset \mathbb{C}$ be open and $f \in C^\infty(D)$. For every $p \in D$ there is a neighborhood $V \subset D$ of p and $u \in C^\infty(V)$ with $\partial u/\partial\bar z = f$ on V.*

Proof. Fix $p \in D$ and choose $r > 0$ with $\overline{B(p, 2r)} \subset D$, together with a cutoff function $\chi \in C_c^\infty(D)$ satisfying $\chi \equiv 1$ on $B(p, r)$ and $\operatorname{supp}(\chi) \subset B(p, 2r)$; set $g := \chi f$, extended by 0 to all of $\mathbb{C}$, so that $g \in C_c^\infty(\mathbb{C})$ and $g = f$ on $B(p, r)$, and it therefore suffices to produce $u \in C^\infty(\mathbb{C})$ with $\partial u/\partial\bar z = g$ on $\mathbb{C}$ and then take $V := B(p, r)$.

Define

$$u(z) := \frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{g(\zeta)}{\zeta - z}\, d\zeta \wedge d\bar\zeta.$$

This is well defined for every z: since g has compact support and $1/|\zeta - z|$ is locally integrable near $\zeta = z$ (in polar coordinates centred at z, $\iint_{|\zeta - z| < \varepsilon} |\zeta - z|^{-1}\, d\zeta \wedge d\bar\zeta$ is finite), the integral converges absolutely.

Substituting $\zeta = z + w$ gives $u(z) = \frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{g(z+w)}{w}\, dw \wedge d\bar w$, in which the singular kernel $1/w$ no longer depends on z; splitting the domain into $|w| < \delta$ and $|w| \geq \delta$, the same local integrability estimate shows the contribution of $|w| < \delta$ to u and to any z-derivative of the integrand is $O(\delta)$ uniformly on compact sets in z (since g and all its derivatives are bounded), while on $|w| \geq \delta$ the integrand is smooth jointly in (z, w) with compact support in w, so differentiation under the integral sign is justified there; letting $\delta \to 0$ shows $u \in C^\infty(\mathbb{C})$, with

$$\frac{\partial u}{\partial \bar z}(z) = \frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{(\partial_{\bar\zeta} g)(\zeta)}{\zeta - z}\, d\zeta \wedge d\bar\zeta$$

obtained by differentiating under the integral sign and substituting back $\zeta = z + w$.

Fix $z \in \mathbb{C}$ and choose $R > 0$ with $\operatorname{supp}(g) \subset B(z, R)$, and for $0 < \varepsilon < R$ let $A_\varepsilon := \{\zeta : \varepsilon \leq |\zeta - z| \leq R\}$ and $\eta := \dfrac{g(\zeta)}{\zeta - z}\, d\zeta$, a $(1,0)$-form on $\mathbb{C}\setminus\{z\}$; by I.5.18, $d\eta = \bar\partial\eta$,

and since $1/(\zeta - z)$ is holomorphic away from $\zeta = z$ this gives $d\eta = \dfrac{(\partial_{\bar\zeta} g)(\zeta)}{\zeta - z}\, d\bar\zeta \wedge d\zeta$, so Stokes' theorem applied to A_ε (I.5.17) yields

$$\iint_{A_\varepsilon} \frac{(\partial_{\bar\zeta} g)(\zeta)}{\zeta - z}\, d\bar\zeta \wedge d\zeta = \int_{|\zeta - z| = R} \frac{g(\zeta)}{\zeta - z}\, d\zeta - \int_{|\zeta - z| = \varepsilon} \frac{g(\zeta)}{\zeta - z}\, d\zeta.$$

The first boundary term vanishes since $g \equiv 0$ on $\{|\zeta - z| = R\}$, while parametrising the second by $\zeta = z + \varepsilon e^{i\theta}$ gives $\int_{|\zeta - z| = \varepsilon} \frac{g(\zeta)}{\zeta - z}\, d\zeta = i \int_0^{2\pi} g(z + \varepsilon e^{i\theta})\, d\theta \to 2\pi i\, g(z)$ as $\varepsilon \to 0$ by continuity of g; letting $\varepsilon \to 0$ on the left-hand side, dominated convergence (again using local integrability of $1/|\zeta - z|$, together with $\partial_{\bar\zeta} g$ bounded and compactly supported) gives

$$\iint_{\mathbb{C}} \frac{(\partial_{\bar\zeta} g)(\zeta)}{\zeta - z}\, d\bar\zeta \wedge d\zeta = -2\pi i\, g(z), \qquad \text{i.e.} \qquad \iint_{\mathbb{C}} \frac{(\partial_{\bar\zeta} g)(\zeta)}{\zeta - z}\, d\zeta \wedge d\bar\zeta = 2\pi i\, g(z).$$

Combining this with the expression for $\partial u/\partial \bar z$ above gives $\dfrac{\partial u}{\partial \bar z}(z) = \dfrac{1}{2\pi i} \cdot 2\pi i\, g(z) = g(z)$ for every $z \in \mathbb{C}$, which restricted to $V = B(p, r)$, where $g = f$, is exactly the claim. □

II. Covering Theory

The main goal of this section is to describe the local behavior of holomorphic maps between Riemann surfaces, using covering space theory. The end of the section is devoted to the proofs of the Riemann–Hurwitz formula and Hurwitz's theorem. The brief discussion of triangulations is intentionally formal and somewhat vague, since a fully detailed treatment would be long and not directly relevant to the purpose of this text.

Throughout this section, X and Y denote Riemann surfaces.

II.1. Topological Covering Spaces

In this subsection we recall the basic notions of covering theory in the topological setting, which will later serve as a model for holomorphic coverings between Riemann surfaces.

II.1.1. Definition [Paths, loops, homotopy] Let V be a topological space, and fix a base point $q \in V$.

A *path* on V is a continuous map $\gamma : [0,1] \to V$.

A *loop based at* q is a path satisfying $\gamma(0) = \gamma(1) = q$.

Two loops γ_1, γ_2 based at q are *homotopic* if there exists a continuous map $G : [0,1] \times [0,1] \to V$ such that $G(0,t) = \gamma_1(t)$, $G(1,t) = \gamma_2(t)$ for all t, and $G(s,0) = G(s,1) = q$ for all s.

II.1.2. Definition [Fundamental group] Homotopy of loops based at q is an equivalence relation. The set of equivalence classes, equipped with concatenation of loops, forms a group called the *fundamental group* of V at q, denoted $\pi_1(V,q)$.

II.1.3. Definition [Covering space] A *covering space* of V is a continuous map $F : U \to V$ such that F is onto, and for every $v \in V$ there exists a neighborhood W of v for which $F^{-1}(W)$ is a disjoint union of open sets U_α, each mapped homeomorphically onto W by F.

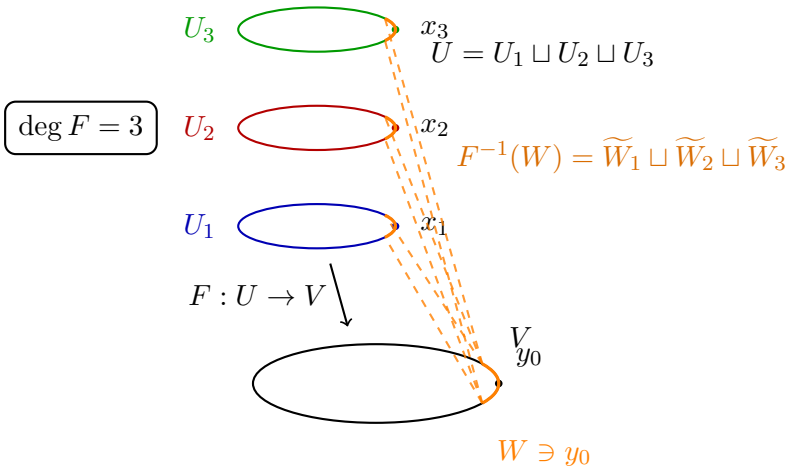


Figure 1: A degree-3 covering $F : U \to V$. The total space U is a disjoint union of three circles U_1, U_2, U_3, each wrapping once around the base circle Y. The base point $y_0 \in Y$ has three preimages x_1, x_2, x_3, one on each sheet. The open arc $W \ni y_0$ is evenly covered: its preimage splits as the disjoint union $F^{-1}(W) = \widetilde{W}_1 \sqcup \widetilde{W}_2 \sqcup \widetilde{W}_3$, with each $\widetilde{W}_i \subset U_i$ mapped homeomorphically onto W by F.

II.1.4. Proposition [Path lifting] *Let $F : U \to V$ be a covering map. For any path $\gamma : [0,1] \to V$ and any $p \in F^{-1}(\gamma(0))$, there exists a unique path $\tilde{\gamma} : [0,1] \to U$ such that $\tilde{\gamma}(0) = p$ and $F \circ \tilde{\gamma} = \gamma$.*

Proof. Uniqueness : if $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ are two lifts with the same initial point, the set

$$S = \{\, t \in [0,1] \mid \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t) \,\}$$

is nonempty, closed, and open in $[0,1]$ because F is a local homeomorphism. Thus $S = [0,1]$, and the lift is unique.

Existence : Define

$$T = \sup\{\, t \in [0,1] \mid \exists\, \tilde{\gamma}_t : [0,t] \to U \text{ with } \tilde{\gamma}_t(0) = p,\ F \circ \tilde{\gamma}_t = \gamma|_{[0,t]} \,\}.$$

Assume $T < 1$. For each $t < T$, let $\tilde{\gamma}_t$ be the unique lift on $[0,t]$. If $t_1 < t_2 < T$, uniqueness implies that $\tilde{\gamma}_{t_2}|_{[0,t_1]} = \tilde{\gamma}_{t_1}$. Thus the family $(\tilde{\gamma}_t)_{t<T}$ is compatible and defines a continuous map

$$\tilde{\gamma} : [0, T[\to U$$

with $F \circ \tilde{\gamma} = \gamma|_{[0,T[}$.

Let $v_0 = \gamma(T)$. Choose an evenly covered neighborhood W of v_0, so that

$$F^{-1}(W) = \bigsqcup_{\alpha} W_\alpha,$$

each $F|_{W_\alpha} : W_\alpha \to W$ being a homeomorphism. Since γ is continuous and $T < 1$, there exists $\varepsilon > 0$ such that $\gamma([T-\varepsilon, T+\varepsilon]) \subset W$.

For t sufficiently close to T, the point $\tilde{\gamma}(t)$ lies in a single sheet W_α (the connectedness of $]T-\varepsilon, T[$ ensures the sheet does not change). Define

$$\tilde{\gamma}(t) = (F|_{W_\alpha})^{-1}(\gamma(t))$$

for $t \in [T, T+\varepsilon]$. This gives a continuous extension of $\tilde{\gamma}$ to $[0, T+\varepsilon]$ satisfying $F \circ \tilde{\gamma} = \gamma|_{[0,T+\varepsilon]}$, contradicting the maximality of T. Hence $T = 1$, and it is a maximum and not only a supremum, since the previous reasoning can be adapted. □

II.1.5. Definition [Isomorphism of coverings] Two coverings $F_1 : U_1 \to V$ and $F_2 : U_2 \to V$ are *isomorphic* if there exists a homeomorphism $G : U_1 \to U_2$ such that $F_2 \circ G = F_1$.

II.1.6. Proposition [Universal covering] *Every connected manifold V admits a covering*

$$F_0 : U_0 \to V$$

with U_0 simply connected. This universal cover *is unique up to isomorphism, and any other connected covering $F : U \to V$ factors uniquely through F_0.*

Proof. Fix a base point $v_0 \in V$. Define

$$U_0 := \big\{[\gamma] : \gamma \text{ is a path in } V \text{ with } \gamma(0) = v_0\big\},$$

where $[\gamma]$ denotes the homotopy class of γ relative to its endpoints, and set $F_0([\gamma]) := \gamma(1)$.

Topology on U_0. For each open simply connected set $W \subset V$ and path γ ending in W, define the basic open set

$$(\gamma, W) := \big\{[\gamma \cdot \delta] : \delta \text{ is a path in } W \text{ with } \delta(0) = \gamma(1)\big\}.$$

These sets form a basis for a topology on U_0. With this topology, $F_0 : U_0 \to V$ is continuous and U_0 is path-connected.

F_0 is a covering map. Let $W \subset V$ be an open simply connected neighborhood. Then

$$F_0^{-1}(W) = \bigsqcup_{[\gamma]} (\gamma, W),$$

where the disjoint union runs over homotopy classes of paths from v_0 to points of W, one for each class. Each restriction $F_0\big|_{(\gamma,W)} : (\gamma, W) \to W$ is a homeomorphism, since W is simply connected (any two paths in W with the same endpoints are homotopic). Hence W is evenly covered.

U_0 is simply connected. Path-connectedness of U_0 follows from the construction. Let $\tilde{\alpha} : [0,1] \to U_0$ be a loop based at $[v_0]$, the class of the constant path at v_0. Set $\alpha := F_0 \circ \tilde{\alpha}$; this is a loop in V based at v_0. The standard lifting of the homotopy contracting α to v_0 lifts to a homotopy in U_0 contracting $\tilde{\alpha}$ to $[v_0]$. Hence $\pi_1(U_0) = 1$.

Factorisation through F_0. Let $F : U \to V$ be a connected covering, and fix $u_0 \in U$ with $F(u_0) = v_0$. For any $[\gamma] \in U_0$, lift γ uniquely to a path $\tilde{\gamma}$ in U starting at u_0 (path-lifting exists and is unique for coverings, by II.1.4); set $\Phi([\gamma]) := \tilde{\gamma}(1)$. This is well-defined: if $\gamma' \cong_{\{0,1\}} \gamma$, the homotopy lifting property gives $\tilde{\gamma}'(1) = \tilde{\gamma}(1)$. One checks directly that Φ is continuous and satisfies $F \circ \Phi = F_0$.

Uniqueness of U_0. If $F_0' : U_0' \to V$ is another universal cover with base point u_0' over v_0, Step 4 applied twice yields continuous maps $\Phi : U_0 \to U_0'$ and $\Psi : U_0' \to U_0$ with $F_0' \circ \Phi = F_0$ and $F_0 \circ \Psi = F_0'$. Both $\Psi \circ \Phi$ and Id_{U_0} are lifts of F_0 agreeing at $[\mathbf{v}_0]$; by uniqueness of lifts, $\Psi \circ \Phi = \mathrm{Id}_{U_0}$, and symmetrically $\Phi \circ \Psi = \mathrm{Id}_{U_0'}$. Hence Φ is an isomorphism of coverings. □

II.2. Covering (Deck) transformations

II.2.1. Definition Let $F : U \to V$ be a covering, denote by $\mathrm{Aut}(X_{/Y})$, the set of automorphisms of U that permutes the fibers of U

$$\{\phi \in \mathrm{Aut}(X), f \circ \phi = f\}.$$

II.2.2. Definition $F : U \to V$ is a Galois covering iif it acts transitively on each fiber, i.e.

$$\forall x, y \in U, (f(x) = f(y)) \implies (\exists \phi \in \mathrm{Aut}(X_{/Y}), \phi(x) = y).$$

II.2.3. Theorem *Let $F_0 : U_0 \to V$ be the universal covering of V, it is a Galois covering.*

Proof. Let $x, y \in U_0$ such that $F_0(x) = F_0(y) = z \in U$. Then this defines 2 different covering of V, and since the universal covering is unique up to isomorphism, the exists an isomorphism of covering ϕ such that $\phi(x) = y$. One can then show can ϕ can be considered as an element of Aut. □

Therefore, we can define a group morphism, let $x \in V$, and consider the mapping

$$\begin{array}{rccc} \psi: & \mathrm{Aut}((U_0)_{/V}) & \longrightarrow & \Pi_1(V,x) \\ & \phi & \longmapsto & [\gamma] \end{array}.$$

Conversely, we can define

$$\begin{array}{rccc} \psi^{-1}: & \Pi_1(V,x) & \longrightarrow & \mathrm{Aut}((U_0)_{/V}) \\ & [\gamma] & \longmapsto & [\alpha] \mapsto [\alpha \cdot \gamma] \end{array}.$$

II.2.4. Proposition *Let $F : X \to Y$ be a covering with Y a Riemann surface and X connected. Then there exists a unique structure of Riemann surface on X that makes F holomorphic. Moreover, every $\phi \in \mathrm{Aut}(X_{/Y})$ is holomorphic.*

II.2.5. Proposition *Let $F : X \to Y$ be a covering, then $\mathrm{Aut}(X_{/Y}$ induces a discrete action on X without fixed point (i.e. for all $x \in X, \exists U$ open subset of $X, \phi(U) \cap U = \varnothing$). The quotient $q : X \to Z := X/\mathrm{Aut}(X_{/Y})$ satisfies*

1. *q is a covering,*
2. *$\exists g : Z \to Y$ covering,*
3. *$f = g \circ q$.*

II.2.6. Remark Z is an intermediate covering, and F is Galois if and only if g is an isomorphism.

II.2.7. Proposition *There is a one-to-one correspondance between connected coverings of $F : X \to Y$ and subgroups $G \subset \mathrm{Aut}((Y_0)_{/Y})$. Moreover, f is Galois if and only if G is normal.*

Proof. Let $F_0 : Y_0 \to Y$ denote the universal covering of Y, and let $\Gamma = \mathrm{Aut}((Y_0)_{/Y})$ be its group of deck transformations. We establish the correspondence in two parts.

Part 1: One-to-one correspondence

From subgroups to coverings: Let $G \subset \Gamma$ be a subgroup. By the preceding proposition, Γ acts discretely and freely on Y_0. Because G is a subgroup, its action on Y_0 is also discrete and free. Therefore, the quotient space $X_G := Y_0/G$ is a well-defined connected topological space, and the quotient map $q : Y_0 \to X_G$ is a covering. Because G consists of automorphisms that preserve the fibers of F_0 (i.e., $F_0 \circ g = F_0$ for all $g \in G$), the map F_0 descends to a well-defined connected covering map $F : X_G \to Y$ such that $F_0 = F \circ q$.

From coverings to subgroups: Conversely, let $F : X \to Y$ be a connected covering. Because Y_0 is simply connected and X is connected, the universal property of covering spaces ensures that the universal cover $F_0 : Y_0 \to Y$ factors through X. That is, there exists a covering map $q : Y_0 \to X$ such that $F_0 = F \circ q$. Consequently, Y_0 also serves as the universal covering of X. We define G to be the group of deck transformations of Y_0 over X:

$$G := \mathrm{Aut}((Y_0)_{/X}) = \{\phi \in \mathrm{Aut}(Y_0) \mid q \circ \phi = q\}$$

For any $\phi \in G$, we have $F_0 \circ \phi = F \circ q \circ \phi = F \circ q = F_0$. This demonstrates that every element of G is also a deck transformation of Y_0 over Y, meaning $G \subset \Gamma$. By the construction of quotient topologies for covering spaces, $X \cong Y_0/G$.

Part 2: F is Galois if and only if G is normal

Let $F : X \to Y$ correspond to the subgroup $G \subset \Gamma$, meaning $X \cong Y_0/G$. An element $\sigma \in \Gamma$ induces a well-defined homeomorphism $\bar{\sigma} : X \to X$ (mapping the orbit $G \cdot y$ to $G \cdot \sigma(y)$) if and only if σ normalizes G. That is, for every $g \in G$, $\sigma g \sigma^{-1} \in G$, meaning $\sigma G \sigma^{-1} = G$. Therefore, the group of deck transformations of X over Y is isomorphic to the quotient of the normalizer of G by G:

$$\mathrm{Aut}(X_{/Y}) \cong N_\Gamma(G)/G$$

By definition, F is a Galois covering if and only if $\mathrm{Aut}(X_{/Y})$ acts transitively on the fibers of F. Because Y_0 is the universal cover, it is a Galois covering, meaning Γ acts transitively on the fibers of F_0. For $\mathrm{Aut}(X_{/Y})$ to act transitively on the fibers of F, the action of $N_\Gamma(G)$ on the fibers of Y_0 over Y must be as extensive as the action of the full group Γ. This transitivity holds if and only if $N_\Gamma(G) = \Gamma$, which is equivalent to G being a normal subgroup of Γ (denoted $G \triangleleft \Gamma$). □

II.3. Multiplicity and Ramification

The Local Normal Form shows that every nonconstant holomorphic map behaves locally like a power map $z \mapsto z^m$ in suitable coordinates. This integer m is the fundamental local invariant of the map.

II.3.1. Definition Let $F : X \to Y$ be a nonconstant holomorphic map between Riemann surfaces, and let $p \in X$. The *multiplicity* of F at p, denoted $\mathrm{mult}_p(F)$, is the unique integer $m \geq 1$ given by the Local Normal Form: in suitable charts centered at p and $F(p)$, the map reads $z \mapsto z^m$. A point $p \in X$ is called a *ramification point* of F if $\mathrm{mult}_p(F) \geq 2$, and a point $y \in Y$ is called a *branch point* if $y = F(p)$ for some ramification point p. When $\mathrm{mult}_p(F) = 1$ the map is a local biholomorphism near p, and p is called an *unramified point*.

II.3.2. Definition A nonconstant holomorphic map between compact Riemann surfaces $F : X \to Y$ is called a *ramified covering of degree* d, for some $d \in \mathbb{N}$, if every unramified value $y \in Y$ (i.e. every y that is not a branch point) has exactly d preimages counted without multiplicity.

II.3.3. Proposition *The degree can equivalently be characterised as*

$$d_y(F) := \sum_{p \in F^{-1}(y)} \mathrm{mult}_p(F) = d =: \deg(F) \qquad \text{for every } y \in Y,$$

where the sum counts all preimages with their multiplicities; this quantity is independent of y whenever Y is connected. Note that this is only well-defined when both of the surfaces are compact.

Proof. We show that $y \mapsto d_y(F)$ is locally constant on Y.

Fix $y \in Y$, and let $x_1, \ldots, x_n$ be the points of $F^{-1}(y)$. By the Local Normal Form, for each i there exist neighborhoods U_i of x_i, a neighborhood V of y, and local coordinates z_i on U_i and w on V such that

$$w = F(z_i) = z_i^{m_i}, \qquad m_i = \mathrm{mult}_{x_i}(F).$$

Shrinking the U_i if necessary, we may assume that they are pairwise disjoint. We claim that, after possibly shrinking V,

$$F^{-1}(V) = \bigsqcup_{i=1}^{n} U_i.$$

Suppose not. Then there is a sequence (q_k) in V converging to y and points $p_k \in X \setminus \bigcup_i U_i$ with $F(p_k) = q_k$. Since X is compact, (p_k) has a convergent subsequence, still denoted (p_k), with limit $p \in X$. By continuity of F, $F(p_k) \to F(p) = y$, so p must be one of the x_i, contradicting $p_k \notin \bigcup_i U_i$. This proves the claim.

Now take any $y' \in V$ with $y' \neq y$; in the coordinate w this means $w' \neq 0$ but small. For each i, the equation $z_i^{m_i} = w'$ has exactly m_i distinct solutions in U_i, each of multiplicity 1, so

$$d_{y'}(F) = \sum_{p \in F^{-1}(y')} \operatorname{mult}_p(F) = \sum_{i=1}^{n} m_i = d_y(F).$$

Thus $y \mapsto d_y(F)$ is locally constant on Y, hence constant since Y is connected. □

II.3.4. Remark When F has no ramification points at all, the two definitions coincide and F is an ordinary (unramified) covering map in the topological sense. Ramification points are thus precisely the points where F fails to be a local homeomorphism.

The following proposition shows that ramification is a genuinely exceptional phenomenon: it can only occur at isolated points.

II.3.5. Proposition [Discreteness of ramification] *Let $F : X \to Y$ be a nonconstant holomorphic map. Then the set of ramification points of F is discrete in X, and the set of branch points is discrete in Y. If X is compact, both sets are finite.*

Proof. In a local chart $\phi : U \to V$ centred at p, the map $F \circ \phi^{-1}$ is a nonconstant holomorphic function, and $\operatorname{mult}_p(F) \geq 2$ if and only if its derivative vanishes at $\phi(p)$. The zeros of a nonconstant holomorphic function are isolated (by the identity theorem), so the ramification points form a discrete subset of X. Since F is open (Open Mapping Theorem), the image of a discrete set under an open map has discrete image, so the branch points are discrete in Y. If X is compact, any discrete subset of X is finite, and $F(X)$ is compact, so the branch points are finite as well. □

The multiplicity is closely related to the order of vanishing of meromorphic functions. The following proposition makes this precise.

II.3.6. Proposition [Multiplicity and order] *Let f be a meromorphic function on a Riemann surface X, and let $F : X \to \mathbb{C}_\infty$ be the associated holomorphic map. Then for every $p \in X$:*

(i) if p is not a pole of f, then $\operatorname{mult}_p(F) = \operatorname{ord}_p\big(f - f(p)\big)$;

(ii) if p is a pole of f, then $\operatorname{mult}_p(F) = -\operatorname{ord}_p(f)$.

Proof. Let z be a local coordinate centered at p, so that $z(p) = 0$.

Case (i): p is not a pole. Then $f(p) \in \mathbb{C}$, and in the standard affine chart of $\mathbb{C}_\infty$ we may write

$$f(z) - f(p) = z^m \, u(z),$$

where $m = \operatorname{ord}_p(f - f(p)) \geq 1$ and u is holomorphic with $u(0) \neq 0$. Since u is nonvanishing near p, it admits a holomorphic m-th root $v = u^{1/m}$ near p (for the same reason as in the proof of I.3.8), and the change of coordinate $w = z\,v(z)$ (which is valid by the inverse function theorem, since $v(0) \neq 0$) transforms $f(z) - f(p)$ into w^m. By the Local Normal Form, $\operatorname{mult}_p(F) = m = \operatorname{ord}_p(f - f(p))$.

Case (ii): p is a pole. Set $g := 1/f$; this is holomorphic near p with $g(p) = 0$. Applying case (i) to g:

$$\mathrm{mult}_p(F) = \mathrm{mult}_p(G) = \mathrm{ord}_p\big(g - g(p)\big) = \mathrm{ord}_p(g),$$

where $G : X \to \mathbb{C}_\infty$ is the holomorphic map associated to g, and we used $g(p) = 0$. Since $g = 1/f$ and p is a pole of f:

$$\mathrm{ord}_p(g) = \mathrm{ord}_p(1/f) = -\,\mathrm{ord}_p(f),$$

which gives $\mathrm{mult}_p(F) = -\,\mathrm{ord}_p(f)$. □

II.4. Triangulations and the Euler Characteristic

II.4.1. Definition A *simplicial complex of dimension* 2 is a union of triangles such that the intersection of any two triangles is either empty, a vertex, or an edge.

II.4.2. Definition A *triangulation* of a compact surface X is a homeomorphism between X and a simplicial complex of dimension 2. We denote by V, E, $\mathcal{F}$ the sets of vertices, edges, and triangles of a triangulation respectively (reserving the letter F for holomorphic maps, as elsewhere in this report).

II.4.3. Remark Every edge belongs to exactly two triangles.

II.4.4. Definition The *Euler characteristic* of a triangulation of a compact surface X is the integer

$$\chi(X) := \#V - \#E + \#\mathcal{F}.$$

II.4.5. Theorem *Every compact surface admits a triangulation, and the Euler characteristic is independent of the choice of triangulation.*

Proof. We sketch the three main steps. First, existence: embed X in $\mathbb{R}^N$ via Whitney's theorem and intersect with a sufficiently fine cubic lattice of mesh size ε; near each point, the surface looks like a plane, so the intersection produces a polygonal decomposition which can be subdivided into triangles.

Second, invariance under subdivision: a barycentric subdivision replaces each triangle by six smaller ones, each edge by two, and adds one vertex per edge and one per triangle. The Euler characteristic is unchanged.

Third, any two triangulations admit a common subdivision, so they have the same Euler characteristic. □

The following theorem is central but we will admit it since it is clearly not the goal of the text.

II.4.6. Theorem [Classification of compact Riemann surfaces by genus] *There exists a unique integer $g_{top} \geq 0$, called the topological genus of X, such that:*

- *$g_{top} = 0$ if and only if X is biholomorphic to the Riemann sphere $\mathbb{P}^1(\mathbb{C})$;*
- *$g_{top} = 1$ if and only if X is biholomorphic to a complex torus $\mathbb{C}/\Lambda$ for some lattice $\Lambda \subset \mathbb{C}$.*

Moreover, two compact Riemann surfaces are homeomorphic if and only if they have the same topological genus.

II.4.7. Remark Formally, the topological genus counts the "number of holes" in the compact Riemann surface.

II.4.8. Proposition *Let Σ_g denote the compact orientable surface of topological genus g (a sphere with g handles). Then $\chi(\Sigma_g) = 2 - 2g$.*

Proof. For $g = 0$, the sphere S^2 is triangulated by the tetrahedron: $V = 4$, $E = 6$, $\mathcal{F} = 4$, giving $\chi = 2$.

For general g, we obtain Σ_g from S^2 by removing $2g$ discs and gluing g cylinders. A disc has Euler characteristic 1 (triangulate it with one central vertex, giving $\chi = 4-5+2 = 1$) and a cylinder has Euler characteristic 0. Therefore

$$\chi(\Sigma_g) = \chi(S^2) - 2g\,\chi(D) + g\,\chi(\text{cyl}) = 2 - 2g \cdot 1 + g \cdot 0 = 2 - 2g.$$

□

II.4.9. Theorem [Riemann–Hurwitz formula] *Let $F : X \to Y$ be a ramified covering of degree d between compact Riemann surfaces. Then*

$$\chi(X) = d \cdot \chi(Y) - \sum_{p \in X} (\text{mult}_p(F) - 1),$$

or equivalently, in terms of genera g_X and g_Y,

$$2g_X - 2 = d \cdot (2g_Y - 2) + \sum_{p \in X} (\text{mult}_p(F) - 1).$$

Proof. Let $B = \{y_1, \ldots, y_n\} \subset Y$ be the set of branch points. Choose a triangulation $\mathcal{T}$ of Y such that each y_j is a vertex. Let V, E, $\mathcal{F}$ denote the numbers of vertices, edges and faces of $\mathcal{T}$, so $\chi(Y) = V - E + \mathcal{F}$.

Lifting the triangulation. Pull back $\mathcal{T}$ to a triangulation $\widetilde{\mathcal{T}}$ of X via F. Away from branch points, each simplex has exactly d preimages. Over a branch point y_j, the degree formula gives $\sum_{p \in F^{-1}(y_j)} \text{mult}_p(F) = d$, so the fiber contributes $\#F^{-1}(y_j)$ vertices instead of d.

Counting simplices of $\widetilde{\mathcal{T}}$. There is no issue lifting the faces and the edges, because all branch points were placed at vertices of the triangulation of Y.

- **Faces:** each face of $\mathcal{T}$ lifts to exactly d faces, giving $d \cdot \mathcal{F}$ faces in $\widetilde{\mathcal{T}}$.
- **Edges:** edges lift to d edges each. Hence $\widetilde{\mathcal{T}}$ has $d \cdot E$ edges.
- **Vertices:** there are approximately $d \cdot V$ vertices in the lift, but this counts each branch point $\sum_p \text{mult}_p(F)$ times instead of once, so we must subtract

  $$\sum_p (\text{mult}_p(F) - 1),$$

  the -1 ensures that each ramification point is still counted once. Therefore

  $$\widetilde{V} = d \cdot V - \sum_p (\text{mult}_p(F) - 1).$$

Euler characteristic.

$$\begin{aligned}\chi(X) &= \widetilde{V} - \widetilde{E} + \widetilde{\mathcal{F}} \\ &= \left(d \cdot V - \sum_p (\mathrm{mult}_p(F) - 1)\right) - dE + d\mathcal{F} \\ &= d \cdot (V - E + \mathcal{F}) - \sum_{p \in X} (\mathrm{mult}_p(F) - 1) \\ &= d \cdot \chi(Y) - \sum_{p \in X} (\mathrm{mult}_p(F) - 1).\end{aligned}$$

□

II.4.10. Proposition [Monodromy representation] *Let $F : X \to Y$ be a ramified covering of degree d, with branch locus $B \subset Y$. Fix a base point $y_0 \in Y \setminus B$ and label $F^{-1}(y_0) = \{x_1, \ldots, x_d\}$. Then there is a group homomorphism*

$$\rho : \pi_1(Y \setminus B,\, y_0) \longrightarrow \mathfrak{S}_d,$$

called the monodromy representation *of F, defined by lifting loops. The covering F is determined up to isomorphism by ρ, and the image $\rho(\pi_1(Y \setminus B))$ acts transitively on $\{1, \ldots, d\}$ since X is connected.*

Proof. Set $Y^\circ := Y \setminus B$ and $X^\circ := X \setminus F^{-1}(B)$. Removing the branch points removes all ramification, so $F : X^\circ \to Y^\circ$ is an ordinary covering of degree d, and II.1.4 applies to it.

Definition of ρ. For a loop class $[\gamma] \in \pi_1(Y^\circ, y_0)$, each x_i determines a unique lift $\widetilde{\gamma}_i$ of γ with $\widetilde{\gamma}_i(0) = x_i$. Its endpoint lies again in the fibre, so

$$\widetilde{\gamma}_i(1) = x_{\sigma(i)}$$

for a unique permutation $\sigma \in \mathfrak{S}_d$. Define $\rho([\gamma]) = \sigma$.

ρ is a homomorphism. If γ and γ' are loops based at y_0, the lift of their concatenation starting at x_i is the lift of γ starting at x_i followed by the lift of γ' starting at $x_{\rho([\gamma])(i)}$. Thus

$$\rho([\gamma \cdot \gamma']) = \rho([\gamma']) \circ \rho([\gamma]),$$

which is a homomorphism after adopting the usual convention for loop composition.

Well-definedness. If γ and γ' are homotopic rel. y_0 in Y°, the homotopy lifts to X° by II.1.4, so the endpoint permutation is the same. Hence ρ is well defined on $\pi_1(Y^\circ, y_0)$.

Transitivity. Assume X is connected. Given $x_i, x_j \in F^{-1}(y_0)$, choose a path α in X from x_i to x_j. Since $F^{-1}(B)$ is finite, α may be perturbed to avoid it; thus $F \circ \alpha$ is a loop in Y°. The lift of $F \circ \alpha$ starting at x_i ends at x_j, so $\rho([F \circ \alpha])$ sends i to j. Hence the image of ρ acts transitively on $\{1, \ldots, d\}$.

Conversely, if the image of ρ is transitive, then any two points of the fibre can be joined by a lifted loop. Since X° is connected, this implies X is connected.

Determination of F by ρ. If two coverings F_1, F_2 have the same monodromy representation, choose basepoints in the fibres and define a map by lifting paths from y_0. Equality of monodromy ensures that the lifted endpoints correspond, so the map is well defined, bijective on fibres, and respects the covering structure. Thus F_1 and F_2 are isomorphic coverings. □

II.5. Group Actions on Riemann Surfaces and the Riemann–Hurwitz theorem

In this subsection we consider the action of a group G on X. The goal is to show that the quotient X/G is itself a Riemann surface, to compute the ramification of the quotient map $\pi : X \to X/G$ explicitly in terms of point stabilisers, and then to feed this ramification data into the Riemann–Hurwitz formula (II.4.9) to obtain Hurwitz's bound on the size of G.

II.5.1. Definition The action of G on X is said to be *holomorphic* if each map $g : X \to X$ is a biholomorphism.

II.5.2. Proposition *Suppose that G acts holomorphically and effectively on X, and that for some $p \in X$ the stabilizer*

$$G_p := \{g \in G \mid g \cdot p = p\}$$

is finite and nontrivial. Then G_p is cyclic.

Proof. Fix $p \in X$ with G_p finite and nontrivial. In a local coordinate z centered at p, any $g \in G_p$ satisfies

$$g(z) = \sum_{n=1}^{\infty} a_n(g)\, z^n,$$

and since g is invertible, $a_1(g) \neq 0$.

Consider the group homomorphism $a_1 : G_p \longrightarrow \mathbb{C}^*$ and suppose that $\ker(a_1)$ contains some $g \neq \mathrm{Id}$. Then in local coordinates we may write

$$g(z) = z + az^m \mod z^{m+1},$$

with $a \neq 0$ and $m > 1$.

Since G_p is finite, there exists $k > 1$ such that $g^k = \mathrm{Id}$. By induction on k, one checks that

$$g^k(z) = z + kaz^m \mod z^{m+1}.$$

But $g^k(z) = z$, hence $a = 0$, absurd. Therefore a_1 is injective.

Since G_p is finite and injects into $\mathbb{C}^*$, and finite subgroups of $\mathbb{C}^*$ are cyclic, it follows that G_p is cyclic. □

II.5.3. Proposition *Let G be a finite group acting holomorphically and effectively on a Riemann surface X. Fix a point $p \in X$. Then there exists an open neighborhood U of p such that:*

(i) U is invariant under the stabilizer G_p, i.e. $g \cdot u \in U$ for every $g \in G_p$ and $u \in U$;

(ii) $U \cap (g \cdot U) = \varnothing$ for every $g \notin G_p$;

(iii) the natural map $\alpha : U/G_p \to X/G$, induced by sending a point of U to its orbit, is a homeomorphism onto an open subset of X/G;

(iv) no point of U except p is fixed by any element of G_p.

Proof. Write $G \setminus G_p = \{g_1, \dots, g_n\}$. Since X is Hausdorff, for each i we may choose open neighborhoods V_i of p and W_i of $g_i \cdot p$ with $V_i \cap W_i = \varnothing$. Then $g_i^{-1} \cdot W_i$ is an open neighborhood of p, and we set

$$R_i = V_i \cap (g_i^{-1} \cdot W_i), \qquad R = \bigcap_{i=1}^{n} R_i, \qquad U = \bigcap_{g \in G_p} g \cdot R.$$

Each R_i is an open neighborhood of p, hence so are R and U. Moreover, for $g \in G_p$ the sets $g \cdot R$ appearing in the definition of U are merely permuted, so $g \cdot U = U$. This proves (i).

For (ii), note that

$$R_i \cap (g_i \cdot R_i) \subset V_i \cap W_i = \varnothing,$$

hence $U \cap (g_i \cdot U) = \varnothing$ for each i.

For (iii), the map $\alpha : U/G_p \to X/G$ is injective: if two points of U have the same G-orbit, they differ by an element of G_p by (ii). The map is continuous and open because its composition with the quotient map $U \to U/G_p$ is the restriction of the quotient map $\pi : X \to X/G$, which is continuous and open. Thus α is a homeomorphism onto its image.

Finally, (iv) follows from the fact that points with nontrivial stabilizer form a discrete subset of X; shrinking U if necessary ensures that p is the only such point in U. □

II.5.4. Theorem *Let G be a finite group acting holomorphically and effectively on a Riemann surface X. Then X/G can be endowed with the structure of a Riemann surface. Moreover, the quotient map*

$$\pi : X \longrightarrow X/G$$

is holomorphic of degree $|G|$, and for every point $p \in X$ one has

$$\operatorname{mult}_p(\pi) = |G_p|.$$

Proof. By II.5.3, for each $p \in X$ there exists an open neighborhood U of p such that:

(i) U is G_p-invariant;

(ii) $U \cap (g \cdot U) = \varnothing$ for all $g \notin G_p$;

(iii) the natural map $\alpha : U/G_p \to W \subset X/G$ is a homeomorphism onto an open neighborhood W of $\bar{p} := \pi(p)$.

We construct a holomorphic chart on W.

Case 1: $|G_p| = 1$. Then $G_p = \{e\}$ and π is a local homeomorphism near p. Choose a local coordinate z on U centered at p; since $\pi|_U : U \to W$ is a homeomorphism, we obtain a chart on W by transporting z:

$$\phi : W \longrightarrow \mathbb{C}, \qquad \phi(\bar{x}) = z\big((\pi|_U)^{-1}(\bar{x})\big).$$

This defines a holomorphic coordinate near $\bar{p}$.

Case 2: $m := |G_p| \geq 2$. Choose a local coordinate z on U centered at p. For each $g \in G_p$, the map g is holomorphic and fixes p, so we may view $g(z)$ as a holomorphic function with a simple zero at $z = 0$. Define

$$h(z) = \prod_{g \in G_p} g(z).$$

Then h is holomorphic on U, G_p-invariant (since G_p permutes the factors), and has multiplicity m at p.

Because h is G_p-invariant, it descends to a continuous map

$$\bar{h} : U/G_p \longrightarrow \mathbb{C}$$

such that $h = \bar{h} \circ q$, where $q : U \to U/G_p$ is the (open) quotient map. Since h is open and q is open, $\bar{h}$ is open.

Near p, the holomorphic map h has multiplicity m, hence is m-to-1 in a small neighborhood of p. On the other hand, away from p, the map q is exactly m-to-1 by construction of U. It follows that $\bar{h}$ is injective near $\bar{p}$. Thus $\bar{h}$ is a continuous, open, injective map from U/G_p onto an open subset $V \subset \mathbb{C}$, hence a homeomorphism onto V.

Composing with $\alpha^{-1} : W \to U/G_p$, we obtain a chart

$$\phi : W \xrightarrow{\alpha^{-1}} U/G_p \xrightarrow{\bar{h}} V \subset \mathbb{C}.$$

This defines a holomorphic coordinate near $\bar{p}$.

In both cases we have constructed holomorphic charts on X/G compatible on overlaps, so X/G is a Riemann surface and $\pi : X \to X/G$ is holomorphic. The degree of π is $|G|$, since each fibre consists of a full G-orbit. Finally, by construction, the multiplicity of π at p is exactly $|G_p|$, because locally the quotient is obtained by dividing by G_p, and the local model is given by h, of multiplicity $|G_p|$. This proves the theorem. □

II.5.5. Remark [Ramification of the quotient map] By definition, $p \in X$ is a ramification point of π precisely when $|G_p| \geq 2$, i.e. when p has nontrivial stabilizer. If X is compact, II.3.5 shows there are only finitely many such points.

II.5.6. Proposition [Ramification sum via orbits] *Suppose that X is compact and let G be a finite group acting holomorphically and effectively on X, with quotient map $\pi : X \to X/G$. For a G-orbit $\mathcal{O} \subset X$ and any $p \in \mathcal{O}$, the orbit-stabilizer theorem gives $|\mathcal{O}| \cdot |G_p| = |G|$, so the contribution of $\mathcal{O}$ to the total ramification is*

$$\sum_{q \in \mathcal{O}} (\mathrm{mult}_q(\pi) - 1) = |\mathcal{O}| \cdot (|G_p| - 1) = |G| \cdot \left(1 - \frac{1}{|G_p|}\right).$$

Consequently, if $\bar{q}_1, \ldots, \bar{q}_r \in X/G$ are the branch points of π and $m_i := |G_{p_i}|$ is the common stabilizer order above $\bar{q}_i$, summing this contribution over the (disjoint) ramified orbits gives

$$\sum_{p \in X} (\mathrm{mult}_p(\pi) - 1) = |G| \cdot \sum_{i=1}^{r} \left(1 - \frac{1}{m_i}\right).$$

Proof. For $p, p' \in \mathcal{O}$ with $p' = h \cdot p$, conjugation gives $G_{p'} = hG_ph^{-1}$, so stabilizers along an orbit are conjugate and in particular have the same order; the orbit–stabilizer theorem then gives $|\mathcal{O}| = |G|/|G_p|$. Since $\mathrm{mult}_q(\pi) = |G_p|$ for every $q \in \mathcal{O}$ (II.5.4),

$$\sum_{q \in \mathcal{O}} (\mathrm{mult}_q(\pi) - 1) = |\mathcal{O}| \cdot (|G_p| - 1) = \frac{|G|}{|G_p|} \cdot (|G_p| - 1) = |G| \left(1 - \frac{1}{|G_p|}\right).$$

The ramification points of π are exactly those with nontrivial stabilizer, and they decompose into the disjoint orbits $\mathcal{O}_1, \ldots, \mathcal{O}_r$ lying above the branch points $\bar{q}_1, \ldots, \bar{q}_r$. Summing the orbit contribution above over $i = 1, \ldots, r$ gives the second formula. □

II.5.7. Corollary [Riemann–Hurwitz formula for group actions] Let G be a finite group acting holomorphically and effectively on a compact Riemann surface X of genus g_X. Let $Y := X/G$ have genus g_Y, and let $\bar{q}_1, \dots, \bar{q}_r \in Y$ be the branch points of $\pi : X \to Y$, with $m_i := |G_{p_i}|$ the stabilizer order above $\bar{q}_i$. Then

$$2g_X - 2 = |G| \cdot \left(2g_Y - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{m_i} \right) \right).$$

Proof. By II.5.4, $\pi : X \to Y = X/G$ is a holomorphic map of degree $d = |G|$, so II.4.9 gives

$$2g_X - 2 = d \cdot (2g_Y - 2) + \sum_{p \in X} (\mathrm{mult}_p(\pi) - 1).$$

Substituting $d = |G|$ and the total ramification formula from II.5.6 yields

$$2g_X - 2 = |G| \cdot (2g_Y - 2) + |G| \cdot \sum_{i=1}^{r} \frac{m_i - 1}{m_i} = |G| \cdot \left(2g_Y - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{m_i} \right) \right) \square$$

II.5.8. Theorem [Hurwitz's automorphism theorem] *Let G be a finite group acting holomorphically and effectively on X of genus $g \geq 2$. Then*

$$|G| \leq 84(g - 1).$$

Proof. Let $Y = X/G$. Since $g \geq 2$, the Riemann–Hurwitz formula for group actions gives

$$2g - 2 = |G| \cdot \left(2g_Y - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{m_i} \right) \right).$$

The left-hand side is positive since $g \geq 2$, so the quantity in parentheses on the right must be positive. We bound it from below by finding its smallest positive value.

Step 1: lower bound on the bracket. Since each $m_i \geq 2$, we have $(m_i - 1)/m_i \geq 1/2$. The bracket $\mu := 2g_Y - 2 + \sum_{i=1}^{r} (m_i - 1)/m_i$ must be positive. We enumerate the cases with μ smallest:

- $g_Y = 0$, $r = 0$: $\mu = -2 < 0$. Not admissible.
- $g_Y = 0$, $r = 1$: $\mu = -2 + \left(1 - \frac{1}{m_i}\right) < 0$. Not admissible.
- $g_Y = 0$, $r = 2$: $\mu = -2 + \frac{m_1 - 1}{m_1} + \frac{m_2 - 1}{m_2} \leq -2 + 1 + 1 = 0$ (with equality only as $m_i \to \infty$). For finite m_i, $\mu < 0$, so not admissible.
- $g_Y = 0$, $r = 3$, $(m_1, m_2, m_3) = (2, 3, 7)$: $\mu = -2 + \frac{1}{2} + \frac{2}{3} + \frac{6}{7} = \frac{1}{42}$. This is the smallest positive value of μ.
- All other configurations with $g_Y = 0$, $r = 3$ or $g_Y \geq 1$ give $\mu \geq 1/42$, as one verifies by exhaustion.

Step 2: the bound. Since $\mu \geq 1/42$ in all admissible cases,

$$2g - 2 = |G| \cdot \mu \geq \frac{|G|}{42},$$

hence $|G| \leq 42(2g - 2) = 84(g - 1)$. $\square$

II.5.9. Remark In fact there is a stronger version of this theorem that states that the groups of automorphisms (biholomorphic function from X to itself) of X is finite.

III. Divisors and their Applications

In this section, X will always denote a Riemann surface.

III.1. Divisors on Riemann Surfaces

The notion of a divisor provides a unified bookkeeping device for the zeros and poles of meromorphic functions and forms. On a compact Riemann surface, this leads to a powerful calculus that relates analytic data (orders of vanishing) to geometric data (degrees of line bundles).

III.1.1. Definition Let X be a Riemann surface. A *divisor* on X is a formal $\mathbb{Z}$-linear combination of points of X,

$$D = \sum_{p \in X} n_p \cdot p, \qquad n_p \in \mathbb{Z},$$

with the requirement that the support $\operatorname{Supp}(D) := \{p \in X : n_p \neq 0\}$ is a discrete subset of X. If X is compact, this forces $\operatorname{Supp}(D)$ to be finite. The set of all divisors on X forms an abelian group under addition, denoted $\operatorname{Div}(X)$.

III.1.2. Definition The *degree* of a divisor $D = \sum n_p \cdot p$ on a compact Riemann surface is the integer

$$\deg D := \sum_{p \in X} n_p \in \mathbb{Z}.$$

The degree map $\deg : \operatorname{Div}(X) \to \mathbb{Z}$ is a group homomorphism.

III.1.3. Remark Note that the compactness is required to ensure the convergence of the sum : the support becomes finite with this condition.

III.1.4. Definition A divisor D is called *effective* (written $D \geq 0$) if $n_p \geq 0$ for all $p \in X$. Given two divisors D and D', we write $D \geq D'$ if $D - D' \geq 0$, i.e. $n_p \geq n'_p$ for all p.

III.2. Divisors of Functions and Forms

III.2.1. Definition Let f be a nonzero meromorphic function on a Riemann surface X. The *divisor of* f is

$$\operatorname{div}(f) := \sum_{p \in X} \operatorname{ord}_p(f) \cdot p.$$

This is well defined: the set of zeros and poles of f is discrete (and finite when X is compact). A divisor of this form is called a *principal divisor*. The set of principal divisors forms a subgroup

$$\operatorname{Prin}(X) \subset \operatorname{Div}(X).$$

III.2.2. Definition Let ω be a nonzero meromorphic 1-form on X. Its *divisor* is

$$\operatorname{Div}(\omega) := \sum_{p \in X} \operatorname{ord}_p(\omega) \cdot p,$$

where $\operatorname{ord}_p(\omega)$ is the order of the coefficient function in any local chart centred at p. A divisor of this form is called a *canonical divisor* and is denoted K. Any two canonical divisors are linearly equivalent (see III.2.3 below).

III.2.3. Definition Two divisors D and D' on X are *linearly equivalent*, written $D \sim D'$, if $D-D'$ is a principal divisor, i.e. $D-D' = \operatorname{div}(f)$ for some nonzero meromorphic function f. Linear equivalence is an equivalence relation on $\operatorname{Div}(X)$, and the quotient group

$$\operatorname{Cl}(X) := \operatorname{Div}(X)/\operatorname{Prin}(X)$$

is called the *Picard group* of X.

III.2.4. Proposition [Degree of a principal divisor] *Suppose X is compact and let f be a nonzero meromorphic function on X. Then*

$$\deg \operatorname{div}(f) = 0.$$

In particular, the degree map descends to a well-defined homomorphism $\deg : \operatorname{Cl}(X) \to \mathbb{Z}$.

Proof. The meromorphic function f defines a holomorphic map $F : X \to \mathbb{C}_\infty$ of some degree $d \geq 1$ (by I.3.7 and the compactness of X). For any regular value $y \in \mathbb{C}_\infty$, the degree formula gives

$$\sum_{p \in F^{-1}(y)} \operatorname{mult}_p(F) = d.$$

Taking $y = 0$: the preimages of 0 are exactly the zeros of f, and $\operatorname{mult}_p(F) = \operatorname{ord}_p(f)$ at each zero by II.3.6. Hence $\sum_{f(p)=0} \operatorname{ord}_p(f) = d$. Taking $y = \infty$: the preimages of ∞ are the poles of f, and $\operatorname{mult}_p(F) = -\operatorname{ord}_p(f)$ at each pole. Hence $\sum_{f(p)=\infty} \operatorname{ord}_p(f) = -d$. Adding:

$$\deg(f) = \sum_{p \in X} \operatorname{ord}_p(f) = 0.$$

□

III.2.5. Proposition *If X is compact, two linearly equivalent divisors have the same degree.*

Proof. If $D_1 \sim D_2$ then $D_1 - D_2 = \operatorname{div}(f)$ for some meromorphic f, and on a compact Riemann surface $\deg(\operatorname{div}(f)) = 0$. The result follows. □

III.2.6. Proposition *Let X be a compact Riemann surface of topological genus g_{top}, and f a non-constant meromorphic function on X. Consider the meromorphic 1-form ω locally defined as $df = f'(z)dz$. Then the following equation holds*

$$\deg \operatorname{div}(\omega) = 2g_{top} - 2.$$

Proof. Consider f as a holomorphic function $F : X \to \mathbb{C}^\infty$ and let $\eta = dz$, the meromorphic 1-form on $\mathbb{C}^\infty$. Note that η is of order -2 at ∞ and 0 elsewhere. We have :

$\omega = F^*\eta$. Using this, I.5.6, II.3.2 and II.4.9 we get

$$\begin{aligned}
\deg \operatorname{div}(\omega) &= \sum_{p\in X} \operatorname{ord}_p \omega, \\
&= \sum_{p\in X} \operatorname{ord}_p F^*\eta, \\
&= \sum_{p\in X} \left((\operatorname{ord}_{F(p)} \eta + 1) \cdot \operatorname{mult}_p F - 1 \right), \\
&= \sum_{\substack{p\in X \\ F(p)\neq\infty}} (\operatorname{mult}_p F - 1) + \sum_{p\in F^{-1}(\infty)} (-\operatorname{mult}_p F - 1), \\
&= \sum_{p\in X} (\operatorname{mult}_p F - 1) - \sum_{p\in F^{-1}(\infty)} 2 \cdot \operatorname{mult}_p F, \\
&= \chi(\mathbb{C}^\infty) \cdot \deg(F) - \chi(X) - 2\deg(F), \\
&= 2g_{top} - 2.
\end{aligned}$$

□

III.3. Intersection Divisors and Bézout's Theorem

Let $X \subset \mathbb{P}^n$ be a smooth projective curve over $\mathbb{C}$ (equivalently, a compact Riemann surface embedded in projective space). We write homogeneous coordinates on $\mathbb{P}^n$ as $[x_0 : \ldots : x_n]$.

III.3.1. Definition [Intersection divisor of a homogeneous polynomial] Let G be homogeneous of degree e and not identically zero on X. For a point $p \in X$ choose a homogeneous polynomial H of the same degree e with $H(p) \neq 0$ (for instance $H = x_i^e$ for some coordinate x_i nonzero at p). Then the quotient G/H is a well-defined meromorphic function in a neighborhood of p on X. Define

$$\operatorname{div}(G)(p) := \operatorname{ord}_p(G/H).$$

For points $q \in X$ where $G(q) \neq 0$ set $\operatorname{div}(G)(q) = 0$. The resulting assignment $p \mapsto \operatorname{div}(G)(p)$ is a divisor on X, called the *intersection divisor* of G with X.

III.3.2. Remark The use of H is necessary here in order to define a function on X, using G alone would create definition problems due to the homogenization.

III.3.3. Lemma [Well-definedness] *The divisor* $\operatorname{div}(G)$ *does not depend on the choice of the auxiliary polynomial* H.

Proof. If H' is another homogeneous polynomial of degree e with $H'(p) \neq 0$, then

$$\frac{G}{H'} = \frac{G}{H} \cdot \frac{H}{H'}.$$

The function H/H' is holomorphic and non vanishing at p, hence has order 0 at p. Multiplying by such a function does not change the order of vanishing, so $\operatorname{ord}_p(G/H) = \operatorname{ord}_p(G/H')$. The integer assigned to p is thus independent of the choice of H. □

III.3.4. Definition [Hyperplane divisor and degree of a curve] A *hyperplane* in $\mathbb{P}^n$ is given by a nonzero homogeneous linear polynomial L; its intersection divisor $L|_X$ is

called a *hyperplane divisor* on X. The *degree* of X, denoted $\deg(X)$, is defined to be the degree of any hyperplane divisor on X (this is well-defined because any two hyperplane divisors are linearly equivalent and X is compact see III.3.5).

III.3.5. Proposition *Let G_1, G_2 be homogeneous polynomials on $\mathbb{P}^n$ of the same degree and assume neither vanishes identically on X. Then the intersection divisors $\operatorname{div}(G_1)$ and $\operatorname{div}(G_2)$ are linearly equivalent on X. In particular all hyperplane divisors are linearly equivalent.*

Proof. Locally on X the quotients G_1/G_2 are meromorphic functions (the two polynomials have the same homogeneity degree, so the quotient is invariant under scaling of homogeneous coordinates). Thus

$$\operatorname{div}(G_1) - \operatorname{div}(G_2) = \operatorname{div}(G_1/G_2)$$

is principal, so the two divisors are linearly equivalent. □

III.3.6. Corollary If X is compact, any two intersection divisors coming from homogeneous polynomials of the same degree have the same degree.

Proof. Linear equivalent divisors on a compact Riemann surface have equal degree; since the difference of two such intersection divisors is principal, their degrees coincide. □

We now state and prove the classical Bézout formula in the curve case.

III.3.7. Theorem [Bézout's theorem] *Let $X \subset \mathbb{P}^n$ be a smooth projective curve of degree $d = \deg(X)$. Let G be a homogeneous polynomial of degree e which does not vanish identically on X. Then the degree of the intersection divisor of G with X equals de:*

$$\deg \operatorname{div}(G) = \deg(X) \cdot \deg(G) = de.$$

Proof. Choose a linear form L (homogeneous of degree 1) such that L does not vanish identically on X; the divisor $\operatorname{div}(L)$ is a hyperplane divisor and by definition $\deg(\operatorname{div}(L)) = \deg(X) = d$. Consider L^e, a homogeneous polynomial of degree e. By the proposition above, $\operatorname{div}(G)$ and $\operatorname{div}(L^e)$ are linearly equivalent divisors on X, hence by the corollary they have the same degree. But

$$\operatorname{div}(L^e) = e \cdot \operatorname{div}(L),$$

so

$$\deg \operatorname{div}(G) = \deg \operatorname{div}(L^e) = e \cdot \deg \operatorname{div}(L) = e \cdot d,$$

as required. □

III.3.8. Remark The proof above gives the intersection number (degree) in the case of a smooth projective curve. In higher dimensions or in the presence of singularities one must refine the statement (counting multiplicities and components properly); the same basic idea—reducing to hyperplane sections and using linear equivalence—remains central.

III.4. Spaces of Functions and Forms Associated to a Divisor

III.4.1. Definition The *space of meromorphic functions with poles bounded by* D, a divisor on X, denoted $L(D)$, is

$$L(D) = \{f \in \mathcal{M}(X) \mid \operatorname{div}(f) + D \geq 0\}.$$

It is direct from the definition that $L(D)$ is a vector space. We use the convention that $0 \in L(D)$ for every divisor D, even though $\operatorname{div}(0)$ is not a divisor.

If D_1, D_2 are two divisors on X with $D_1 \leq D_2$, then $L(D_1) \subset L(D_2)$. Moreover, a meromorphic function f is holomorphic if and only if $\operatorname{div}(f) \geq 0$, so that $L(0) = \mathcal{O}_X(X)$. If X is compact, then $L(0) \cong \mathbb{C}$ by I.2.5.

III.4.2. Lemma *If X is compact and $D \in \operatorname{Div}(X)$ is a divisor with $\deg D < 0$, then $L(D) = \{0\}$.*

Proof. Let $f \in L(D)$, and suppose for contradiction that $f \neq 0$. By definition of $L(D)$, we have $\operatorname{div}(f) + D \geq 0$, so $\operatorname{div}(f) + D$ is effective. Hence $\deg(\operatorname{div}(f) + D) \geq 0$.

Since X is compact and f is a nonzero meromorphic function, III.2.4 gives $\deg \operatorname{div}(f) = 0$. Therefore,

$$\deg D = \deg \operatorname{div}(f) + \deg D = \deg(\operatorname{div}(f) + D) \geq 0,$$

which contradicts the assumption that $\deg D < 0$. Thus $f = 0$, and hence $L(D) = \{0\}$. □

III.4.3. Proposition *Let D_1, D_2 be two linearly equivalent divisors on X. Then $L(D_1) \cong L(D_2)$. More precisely, if $D_1 = \operatorname{div}(h) + D_2$, multiplication by h induces an isomorphism between these spaces.*

Proof. Let $f \in L(D_1)$. Then $\operatorname{div}(f) + D_1 \geq 0$, so $\operatorname{div}(f) \geq -D_1$. Hence

$$\begin{aligned} \operatorname{div}(hf) &= \operatorname{div}(h) + \operatorname{div}(f), \\ &\geq \operatorname{div}(h) - D_1, \\ &= -D_2. \end{aligned}$$

Thus $hf \in L(D_2)$. Therefore multiplication by h defines a linear map $\mu_h : L(D_1) \to L(D_2)$. Its inverse is multiplication by $1/h$, so μ_h is an isomorphism. □

We have analogous definitions and properties for meromorphic 1-forms.

III.4.4. Definition The *space of meromorphic 1-forms with poles bounded by* D, a divisor on X, denoted $L^{(1)}(D)$, is

$$L^{(1)}(D) = \left\{\omega \in \mathcal{M}^{(1)}(X) \mid \operatorname{div}(\omega) + D \geq 0\right\}.$$

Again, $L^{(1)}(D)$ is a vector space. If $D_1 \leq D_2$, then $L^{(1)}(D_1) \subset L^{(1)}(D_2)$. Moreover, a meromorphic 1-form ω is holomorphic if and only if $\operatorname{div}(\omega) \geq 0$, so $L^{(1)}(0) = \Omega^1_X(X)$.

Let η be a nonzero meromorphic 1-form on X, and let $K = \operatorname{div}(\eta)$ be the associated canonical divisor. Then every meromorphic 1-form ω can be written uniquely as $\omega = f\eta$ for some meromorphic function f.

III.4.5. Proposition *Let $K = \operatorname{div}(\eta)$ be a canonical divisor. Then $L^{(1)}(-D) \cong L(K - D)$. More precisely, the map $f \mapsto f\eta$ induces an isomorphism $L(K - D) \to L^{(1)}(-D)$.*

Proof. For $f \in \mathcal{M}(X)$, one has

$$\operatorname{div}(f\eta) - D = \operatorname{div}(f) + K - D.$$

Thus $f\eta \in L^{(1)}(-D)$ if and only if $f \in L(K - D)$. Since every meromorphic 1-form is uniquely of the form $f\eta$, multiplication by η gives the desired isomorphism. □

III.4.6. Corollary If X is compact, then $\dim L^{(1)}(-D) = \dim L(K - D)$.

III.4.7. Corollary If X is compact and $\deg(K - D) < 0$, then $L^{(1)}(-D) = \{0\}$.

Proof. Clear by III.4.2. □

III.4.8. Proposition *Let D_1, D_2 be two linearly equivalent divisors on X. Then $L^{(1)}(D_1) \cong L^{(1)}(D_2)$. More precisely, if $D_1 = \operatorname{div}(h) + D_2$, multiplication by h induces an isomorphism between these spaces.*

Proof. The proof is the same as in III.4.5. □

IV. Sheaves and Čech Cohomology

We have seen with Liouville's theorem that considering global holomorphic functions on a compact Riemann surface isn't giving us much information. Thus, the local properties carry the information, and we need a tool to take it to the global level. This is exactly what are sheaves meant for.

IV.1. Presheaves and Sheaves

In order to define a local property, we need to define it on all opens of the topological space we're considering.

IV.1.1. Definition Let X be a topological space. A *presheaf of groups* on X is a collection of groups $\mathcal{F}(U)$ for every open set U, and a collection of group homomorphisms

$$\rho_V^U : \mathcal{F}(U) \to \mathcal{F}(V)$$

whenever $V \subset U$ such that :

- $\mathcal{F}(\varnothing)$ is the trivial group,
- $\rho_U^U = \mathrm{Id}$, and
- if $W \subset V \subset U$, then $\rho_V^U \circ \rho_W^V = \rho_W^U$.

The homomorphisms ρ_V^U are called the *restriction maps* of the presheaf. The elements of $\mathcal{F}(U)$ are called the *sections* of $\mathcal{F}$ over U, and the ones of $\mathcal{F}(X)$ are the *global sections* of $\mathcal{F}$.

IV.1.2. Remark One can define a presheaf of rings similarly.

In order to define a sheaf, we have to add a condition on the sections, so that the local properties can lift properly.

IV.1.3. Definition [**The sheaf axiom**] Let U be an open of a topological space X, and $\mathcal{F}$ a presheaf on X. Suppose that $\mathfrak{U} = (U_i)_{i\in I}$ is an open covering of U and that for each U_i we are given a section $s_i \in \mathcal{F}(U_i)$ such that for all $i, j \in I$,

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

We say that $\mathcal{F}$ verifies the *sheaf axiom* if we can patch together all the s_i into a unique global section : there exists a unique $s \in \mathcal{F}(U)$ such that for all $i \in I$,

$$\rho_{U_i}^X(s) = s_i.$$

IV.1.4. Definition The *sheaf of holomorphic functions* on a topological space X, $\mathcal{O}_X$: for U open on X, define

$$\mathcal{O}_X(U) := \{\, \text{holomorphic functions } f : U \to \mathbb{C} \,\}.$$

Note that it verifies the sheaf axiom because of the identity theorem I.2.3.

IV.1.5. Definition Similarly, the *sheaf of meromorphic functions* on a topological space X is $\mathcal{M}_X$ where for U open on X, we define $\mathcal{M}_X(U) := \{\, \text{meromorphic functions } f : U \to \mathbb{C} \,\}$.

IV.1.6. Definition Let X be a Riemann surface, and D a divisor on X. Then

$$\mathcal{O}_X[D](U) := \{f \in \mathcal{M}_X(U),\ \mathrm{div}(f) + D \geq 0 \text{ on } U\}.$$

is a sheaf.

IV.1.7. Definition [**Laurent tail divisors and their sheaf**] Let X be a Riemann surface and D a divisor on X. For each point $p \in X$, fix a local coordinate z_p centered at p, and define the *local Laurent tail group at p bounded by D* as the group of Laurent polynomials

$$\mathcal{T}_p[D] := \left\{ \sum_{n<-D(p)} a_n z_p^n \;\middle|\; a_n \in \mathbb{C},\ \text{finitely many } a_n \neq 0 \right\}.$$

Note that $\mathcal{T}_p[D] = \{0\}$ whenever $D(p) \geq 0$. The *skyscraper sheaf of Laurent tail divisors* $\mathcal{T}_X[D]$ is the sheaf defined by

$$\mathcal{T}_X[D](U) := \prod_{p \in U} \mathcal{T}_p[D]$$

for every open set $U \subset X$. When X is compact, the support $\{p \in X : D(p) < 0\}$ is finite, so the product is finite and the group of global sections reduces to

$$\mathcal{T}_X[D](X) = \bigoplus_{p \in X} \mathcal{T}_p[D].$$

Finally, there is a natural *Laurent tail map*

$$\alpha_D : \mathcal{M}(X) \longrightarrow \mathcal{T}_X[D](X),$$

sending a meromorphic function f to $(r_p)_{p \in X}$, where r_p is the truncation of the Laurent series of f in the coordinate z_p, retaining only terms of degree strictly less than $-D(p)$.

IV.1.8. Definition [**Skyscraper sheaf at a point**] Let X be a topological space and $p \in X$. The *complex skyscraper sheaf at p*, denoted $\mathbb{C}_p$, is the sheaf defined by

$$\mathbb{C}_p(U) := \begin{cases} \mathbb{C} & \text{if } p \in U, \\ \{0\} & \text{if } p \notin U, \end{cases}$$

for every open set $U \subset X$, with restriction maps given by the identity $\mathrm{Id}_{\mathbb{C}}$ when both opens contain p, and by the zero map otherwise.

IV.1.9. Definition [**Sheaf of divisors**] Let X be a Riemann surface. The *sheaf of divisors* $\mathcal{D}iv_X$ is the sheaf defined by

$$\mathcal{D}iv_X(U) := \left\{ \sum_{p \in U} n_p \cdot p \;\middle|\; n_p \in \mathbb{Z},\ \{p \in U : n_p \neq 0\} \text{ is discrete in } U \right\}$$

for every open set $U \subset X$.

IV.2. Sheaf Maps

As usual, defining objects is not enough, and one needs to define maps between them. In the rest of this section, X will denote a topological space.

IV.2.1. Definition Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves of groups on X. A *morphism of presheaves* $\varphi : \mathcal{F} \to \mathcal{G}$ is a collection of group homomorphisms

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U), \qquad U \text{ open in } X,$$

compatible with restrictions, i.e. such that for every open $V \subset U$ and every $s \in \mathcal{F}(U)$,

$$\varphi_V(s|_V) = \varphi_U(s)|_V.$$

If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then φ is called a *morphism of sheaves*, or a *sheaf map*. We say that φ is an *isomorphism* if there exists a morphism $\psi : \mathcal{G} \to \mathcal{F}$ such that $\psi \circ \varphi = \mathrm{Id}_{\mathcal{F}}$ and $\varphi \circ \psi = \mathrm{Id}_{\mathcal{G}}$. Equivalently, φ_U is an isomorphism of groups for every open set $U \subset X$.

IV.2.2. Example The inclusion map defines a morphism of sheaves $\mathcal{O}_X \to \mathcal{M}_X$. More generally, if $D_1 \leq D_2$ are two divisors, the inclusion defines a morphism from $\mathcal{O}_X[D_1]$ to $\mathcal{O}_X[D_2]$.

In order to compare sheaves *locally*, we introduce the following notion.

IV.2.3. Definition Let $\mathcal{F}$ be a presheaf on X and $x \in X$. The *stalk* of $\mathcal{F}$ at x is the group

$$\mathcal{F}_x := \varinjlim_{U \ni x} \mathcal{F}(U),$$

the direct limit taken over the open neighborhoods of x, ordered by reverse inclusion, with transition maps the restriction maps of $\mathcal{F}$. Concretely, $\mathcal{F}_x$ is the set of pairs (U, s), with $U \ni x$ open and $s \in \mathcal{F}(U)$, modulo the relation $(U, s) \sim (V, t)$ if $s|_W = t|_W$ for some open $W \subset U \cap V$ containing x. The class of (U, s) is called the *germ* of s at x, and denoted s_x.

A morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$ induces, for every $x \in X$, a homomorphism on stalks $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$, $s_x \mapsto \varphi_U(s)_x$ for $s \in \mathcal{F}(U)$, $x \in U$.

IV.2.4. Definition A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ is *injective* (resp. *surjective*) if $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective (resp. surjective) for every $x \in X$.

IV.2.5. Proposition *A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ is injective if and only if $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for every open $U \subset X$.*

IV.2.6. Remark The analogous statement for surjectivity is *false*: φ may be surjective without any φ_U being surjective.

The following propositions stem from the definitions of sheaves and stalks of sheaves.

IV.2.7. Proposition *A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ is an isomorphism if and only if $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is bijective for every $x \in X$.*

IV.2.8. Proposition *A morphism of sheaves $\psi : \mathcal{F} \to \mathcal{G}$ is surjective if and only if for every open $U \subset X$, every $t \in \mathcal{G}(U)$ and every $x \in U$, there exist an open neighborhood $V \subset U$ of x and a section $s \in \mathcal{F}(V)$ such that $\psi_V(s) = t|_V$.*

IV.2.9. Definition Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. The *kernel* of φ is the presheaf $\ker \varphi$ defined by

$$(\ker \varphi)(U) := \ker(\varphi_U) \subset \mathcal{F}(U)$$

for every open U, with the restriction maps induced by those of $\mathcal{F}$.

IV.2.10. Proposition *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then* $\ker \varphi$ *is a sheaf, and for every $x \in X$ one has*

$$(\ker \varphi)_x = \ker(\varphi_x).$$

IV.2.11. Remark A sheaf $\mathcal{H}$ with $\mathcal{H}(U) \subset \mathcal{F}(U)$ for every open U, compatibly with restrictions, is called a *subsheaf* of $\mathcal{F}$. For instance $\ker \varphi$ is a subsheaf of $\mathcal{F}$, and so is $\mathcal{O}_X[D] \subset \mathcal{O}_X$.

IV.3. Exact Sequences of Sheaves

Many sheaves of interest fit into exact sequences relating them to simpler ones.

IV.3.1. Definition A sequence of sheaf morphisms

$$\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^{i} \xrightarrow{\varphi^{i}} \mathcal{F}^{i+1} \to \cdots$$

is *exact at* $\mathcal{F}^i$ if $\operatorname{im}(\varphi^{i-1}_x) = \ker(\varphi^i_x)$ for every $x \in X$; it is *exact* if it is exact at every term. A *short exact sequence* is an exact sequence of the form

$$0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0,$$

where 0 denotes the sheaf with $0(U) = \{0\}$ for every open U. Concretely, this means that φ is injective, ψ is surjective, and the sequence of stalks $0 \to \mathcal{F}'_x \to \mathcal{F}_x \to \mathcal{F}''_x \to 0$ is exact for every $x \in X$.

IV.3.2. Proposition *Let $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ be a short exact sequence of sheaves on X. Then for every open $U \subset X$, the sequence of groups*

$$0 \to \mathcal{F}'(U) \xrightarrow{\varphi_U} \mathcal{F}(U) \xrightarrow{\psi_U} \mathcal{F}''(U)$$

is exact. One says that the functor $\mathcal{F} \mapsto \mathcal{F}(U)$ is left exact.

Proof. Injectivity of φ_U follows from injectivity on stalks: if $\varphi_U(s) = 0$, then for every $x \in U$ the germ $\varphi_x(s_x) = 0$, hence $s_x = 0$ because φ_x is injective. Thus $s = 0$ by the sheaf axiom.

Next, $\psi_U \circ \varphi_U = 0$ holds because $\psi \circ \varphi = 0$ as sheaf morphisms, so $\operatorname{im}(\varphi_U) \subseteq \ker(\psi_U)$.

For the reverse inclusion, let $t \in \mathcal{F}(U)$ satisfy $\psi_U(t) = 0$. For each $x \in U$, the germ t_x lies in $\ker(\psi_x)$, and exactness of

$$0 \to \mathcal{F}'_x \to \mathcal{F}_x \to \mathcal{F}''_x \to 0$$

gives a unique $s_x \in \mathcal{F}'_x$ with $\varphi_x(s_x) = t_x$. By definition of stalks, for each x there exists an open $U_x \subseteq U$ and a section $s'_x \in \mathcal{F}'(U_x)$ representing s_x.

On overlaps $U_x \cap U_y$, the germs of s'_x and s'_y coincide, because both map to t under φ. Since $\mathcal{F}'$ is a sheaf, the s'_x glue to a unique $s \in \mathcal{F}'(U)$ with $\varphi_U(s) = t$. Thus $\ker(\psi_U) \subseteq \operatorname{im}(\varphi_U)$.

Exactness follows. □

IV.3.3. Remark The map ψ_U above need not be surjective: a section of $\mathcal{F}''$ over U may only lift to a section of $\mathcal{F}$ *locally* near each point of U (IV.2.8), with no guarantee that the local lifts patch into a global one. Measuring this obstruction is exactly the role of the cohomology groups constructed below.

IV.3.4. Example Let X be a Riemann surface, D a divisor on X, and $p \in X$. Fix a local coordinate z centered at p: for $U \ni p$ open and $f \in \mathcal{O}_X[D](U)$, write $f = \sum_{n \geq D(p)} a_n z^n$ and set $\mathrm{ev}_p(f) := a_{D(p)}$ (and $\mathrm{ev}_p(f) := 0$ if $p \notin U$). This defines a morphism of sheaves $\mathrm{ev}_p : \mathcal{O}_X[D] \to \mathbb{C}_p$, where $\mathbb{C}_p$ is the skyscraper sheaf with group $\mathbb{C}$, fitting into a short exact sequence

$$0 \to \mathcal{O}_X[D-p] \to \mathcal{O}_X[D] \xrightarrow{\mathrm{ev}_p} \mathbb{C}_p \to 0.$$

For $D = 0$ this is the evaluation sequence $0 \to \mathcal{O}_X[-p] \to \mathcal{O}_X \xrightarrow{f \mapsto f(p)} \mathbb{C}_p \to 0$. By IV.3.2, taking global sections gives an exact sequence

$$0 \to \mathcal{O}_X[D-p](X) \to \mathcal{O}_X[D](X) \to \mathbb{C},$$

but the last map need not be onto: this failure is one of the starting points of the Riemann–Roch theorem.

IV.3.5. Remark Another classical instance of this phenomenon is the *exponential sequence* $0 \to \underline{\mathbb{Z}}_X \to \mathcal{O}_X \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_X^* \to 0$, where $\underline{\mathbb{Z}}_X$ is the locally constant sheaf with values in $\mathbb{Z}$ and $\mathcal{O}_X^*$ is the sheaf of non-vanishing holomorphic functions: a non-vanishing holomorphic function need not have a global holomorphic logarithm.

IV.4. Čech Cohomology

We can now construct the tool announced at the beginning of this section: a cohomology theory measuring the obstruction described in IV.3.2. From now on, all (pre)sheaves are sheaves of *abelian* groups, so that the alternating sums below make sense (this holds in every example seen so far, taking the group G in the skyscraper sheaf to be abelian).

IV.4.1. Definition Let $\mathcal{F}$ be a sheaf of abelian groups on X and $\mathfrak{U} = (U_i)_{i \in I}$ an open covering of X. For $i_0, \ldots, i_q \in I$, write $U_{i_0 \cdots i_q} := U_{i_0} \cap \cdots \cap U_{i_q}$. For $q \geq 0$, the group of *Čech q-cochains* of $\mathcal{F}$ relative to $\mathfrak{U}$ is

$$C^q(\mathfrak{U}, \mathcal{F}) := \prod_{(i_0, \ldots, i_q) \in I^{q+1}} \mathcal{F}(U_{i_0 \cdots i_q})$$

(a factor with $U_{i_0 \cdots i_q} = \varnothing$ contributes $\mathcal{F}(\varnothing) = \{0\}$). We write $\sigma = (\sigma_{i_0 \cdots i_q})$ for an element of $C^q(\mathfrak{U}, \mathcal{F})$. The *coboundary operator* $d = d^q : C^q(\mathfrak{U}, \mathcal{F}) \to C^{q+1}(\mathfrak{U}, \mathcal{F})$ is defined by

$$(d\sigma)_{i_0 \cdots i_{q+1}} := \sum_{k=0}^{q+1} (-1)^k \, \sigma_{i_0 \cdots \widehat{i_k} \cdots i_{q+1}} \Big|_{U_{i_0 \cdots i_{q+1}}},$$

where $\widehat{i_k}$ means that the index i_k is omitted.

IV.4.2. Proposition *$d^2 = 0$ for every $q \geq 0$.*

Proof. Let $\sigma \in C^{q-1}(\mathfrak{U}, \mathcal{F})$, we compute

$$\begin{aligned}
(d^q d^{q-1}\sigma)_{i_0, \ldots, i_{q+1}} &= \sum_{\ell=0}^{q+1} (-1)^\ell \, (d^{q-1}\sigma)_{i_0, \ldots, \widehat{i_\ell}, \ldots, i_{q+1}} \\
&= \sum_{\ell=0}^{q+1} (-1)^\ell \left[\sum_{k=0}^{\ell-1} (-1)^k \sigma_{i_0, \ldots, \widehat{i_k}, \ldots, \widehat{i_\ell}, \ldots, i_{q+1}} + \sum_{k=\ell+1}^{q+1} (-1)^{k-1} \sigma_{i_0, \ldots, \widehat{i_\ell}, \ldots, \widehat{i_k}, \ldots, i_{q+1}} \right]
\end{aligned}$$

Each term $c_{i_0, \ldots, \widehat{i_\ell}, \ldots, \widehat{i_k}, \ldots, i_{q+1}}$ appears twice with opposite signs. Thus all terms cancel pairwise, and $d^2 = 0$. □

IV.4.3. Definition The group of *q-cocycles* is $Z^q(\mathfrak{U},\mathcal{F}) := \ker d^q$, and the group of *q-coboundaries* is $B^q(\mathfrak{U},\mathcal{F}) := \operatorname{im} d^{q-1}$ for $q \geq 1$ (with $B^0(\mathfrak{U},\mathcal{F}) := 0$). By the previous proposition, $B^q(\mathfrak{U},\mathcal{F}) \subset Z^q(\mathfrak{U},\mathcal{F})$, and we define the *q-th Čech cohomology group of* $\mathcal{F}$ *relative to* $\mathfrak{U}$ as

$$H^q(\mathfrak{U},\mathcal{F}) := Z^q(\mathfrak{U},\mathcal{F})/B^q(\mathfrak{U},\mathcal{F}).$$

IV.4.4. Proposition *For every open covering* $\mathfrak{U} = (U_i)_{i\in I}$ *of* X*, the map* $\mathcal{F}(X) \to C^0(\mathfrak{U},\mathcal{F})$*,* $s \mapsto (s|_{U_i})_{i\in I}$*, induces an isomorphism*

$$\mathcal{F}(X) \cong H^0(\mathfrak{U},\mathcal{F}).$$

Proof. Let

$$\begin{array}{rccc} \alpha: & \mathcal{F}(X) & \longrightarrow & C^0(\mathfrak{U},\mathcal{F}) \\ & f & \longmapsto & (f_i) \end{array}.$$

Since $d(f_i) = (f_j - f_i)$, the map α lands in $Z^0(\mathfrak{U},\mathcal{F})$, and hence in $H^0(\mathfrak{U},\mathcal{F})$ since there are no non-zero 0-coboundaries. Thus, α is injective and surjective by the sheaf axiom. □

In order to get rid of the choice of $\mathfrak{U}$, we let the covering vary.

IV.4.5. Definition Let $\mathfrak{U} = (U_i)_{i\in I}$ and $\mathfrak{V} = (V_j)_{j\in J}$ be open coverings of X. We say $\mathfrak{V}$ *refines* $\mathfrak{U}$ if there is a map $r : J \to I$ with $V_j \subset U_{r(j)}$ for every $j \in J$. Given such a r, the *refinement map* $r^* : C^q(\mathfrak{U},\mathcal{F}) \to C^q(\mathfrak{V},\mathcal{F})$ is defined by

$$(r^*\sigma)_{j_0\cdots j_q} := \sigma_{r(j_0)\cdots r(j_q)}\Big|_{V_{j_0\cdots j_q}}.$$

IV.4.6. Proposition *The map* r^* *satisfies* $r^* \circ d = d \circ r^*$*, hence induces a homomorphism, still denoted* $r^* : H^q(\mathfrak{U},\mathcal{F}) \to H^q(\mathfrak{V},\mathcal{F})$*, which does not depend on the choice of* r*.*

Proof. Let $\sigma \in C^q(\mathfrak{U},\mathcal{F})$. By definition of the Čech differential d, we have

$$\begin{aligned} (d(r^*\sigma))_{j_0\cdots j_{q+1}} &= \sum_{m=0}^{q+1} (-1)^m (r^*\sigma)_{j_0\cdots\widehat{j_m}\cdots j_{q+1}} \\ &= \sum_{m=0}^{q+1} (-1)^m \sigma_{r(j_0)\cdots\widehat{r(j_m)}\cdots r(j_{q+1})} \\ &= (d\sigma)_{r(j_0)\cdots r(j_{q+1})} \\ &= (r^*(d\sigma))_{j_0\cdots j_{q+1}}. \end{aligned}$$

Thus $r^* \circ d = d \circ r^*$, so r^* induces a homomorphism on cohomology.

To show independence of r, let $r, r' : J \to I$ be two refinement maps. Define a homotopy operator $k : C^q(\mathfrak{U},\mathcal{F}) \to C^{q-1}(\mathfrak{V},\mathcal{F})$ by

$$(k\sigma)_{j_0\cdots j_{q-1}} = \sum_{m=0}^{q-1} (-1)^m \sigma_{r(j_0)\cdots r(j_m) r'(j_m)\cdots r'(j_{q-1})}.$$

Let $\sigma \in C^q(\mathfrak{U}, \mathcal{F})$. First, we compute $(kd\sigma)_{j_0 \cdots j_q}$ by applying the definition of k to $d\sigma$:

$$\begin{aligned}(kd\sigma)_{j_0\cdots j_q} &= \sum_{m=0}^{q}(-1)^m(d\sigma)_{r(j_0)\cdots r(j_m)r'(j_m)\cdots r'(j_q)} \\ &= \sum_{m=0}^{q}(-1)^m\left(\sum_{n=0}^{m}(-1)^n\sigma_{r(j_0)\cdots\widehat{r(j_n)}\cdots r(j_m)r'(j_m)\cdots r'(j_q)}\right. \\ &\qquad \left. + \sum_{n=m}^{q}(-1)^{n+1}\sigma_{r(j_0)\cdots r(j_m)r'(j_m)\cdots\widehat{r'(j_n)}\cdots r'(j_q)}\right).\end{aligned}$$

Then we compute $(dk\sigma)_{j_0 \cdots j_q}$ by applying the definition of d to $k\sigma$:

$$\begin{aligned}(dk\sigma)_{j_0\cdots j_q} &= \sum_{n=0}^{q}(-1)^n(k\sigma)_{j_0\cdots\widehat{j_n}\cdots j_q} \\ &= \sum_{n=0}^{q}(-1)^n\left(\sum_{m=0}^{n-1}(-1)^m\sigma_{r(j_0)\cdots r(j_m)r'(j_m)\cdots\widehat{r'(j_n)}\cdots r'(j_q)}\right. \\ &\qquad \left. + \sum_{m=n+1}^{q}(-1)^{m-1}\sigma_{r(j_0)\cdots\widehat{r(j_n)}\cdots r(j_m)r'(j_m)\cdots r'(j_q)}\right), \\ &= \sum_{m=0}^{q}(-1)^m\left(\sum_{n=0}^{m-1}(-1)^{n-1}\sigma_{r(j_0)\cdots\widehat{r(j_n)}\cdots r(j_m)r'(j_m)\cdots r'(j_q)}\right. \\ &\qquad \left. + \sum_{n=m+1}^{q}(-1)^{n}\sigma_{r(j_0)\cdots r(j_m)r'(j_m)\cdots\widehat{r'(j_n)}\cdots r'(j_q)}\right).\end{aligned}$$

Comparing the two double sums, the terms where $k \neq m$ in $kd\sigma$ cancel exactly with the terms in $dk\sigma$ because their signs are opposite.

Thus:

$$\begin{aligned}(kd\sigma)_{j_0\cdots j_q} + (dk\sigma)_{j_0\cdots j_q} &= \sum_{m=0}^{q}\sigma_{r(j_0)\cdots r(j_{m-1})r'(j_m)\cdots r'(j_q)} - \sigma_{r(j_0)\cdots r(j_m)r'(j_{m+1})\cdots r'(j_q)}, \\ &= \sigma_{r'(j_0)\cdots r'(j_q)} - \sigma_{r(j_0)\cdots r(j_q)} \\ &= (r'^*\sigma)_{j_0\cdots j_q} - (r^*\sigma)_{j_0\cdots j_q}.\end{aligned}$$

Thus, $r'^* - r^* = dk + kd$ and therefore, r^* and r'^* induce the same map on $H^q(\mathfrak{U}, \mathcal{F})$ since kd is zero on $Z^q(\mathfrak{U}, \mathcal{F})$ and dk is a q-coboundary. □

IV.4.7. Definition The open coverings of X are directed by refinement: any two coverings $\mathfrak{U}, \mathfrak{V}$ admit the common refinement $\mathfrak{U} \wedge \mathfrak{V} := (U_i \cap V_j)_{(i,j)\in I\times J}$. By IV.4.6, the groups $H^q(\mathfrak{U}, \mathcal{F})$, together with the refinement maps, form a direct system of abelian groups indexed by the open coverings of X. The *q-th Čech cohomology group of X with values in $\mathcal{F}$* is

$$H^q(X, \mathcal{F}) := \varinjlim_{\mathfrak{U}} H^q(\mathfrak{U}, \mathcal{F}).$$

The following proposition is directly deduced from IV.4.4.

IV.4.8. Proposition $H^0(X, \mathcal{F}) \cong \mathcal{F}(X)$.

IV.4.9. Proposition *A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ induces, for every $q \geq 0$, a homomorphism $\varphi_* : H^q(X, \mathcal{F}) \to H^q(X, \mathcal{G})$, obtained by applying φ termwise to cochains over any covering (this commutes with d and with refinement maps). Moreover $(\mathrm{Id}_{\mathcal{F}})_* = \mathrm{Id}$ and $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$: that is, $H^q(X, -)$ is a functor from sheaves of abelian groups on X to abelian groups.*

Proof. This can be deduced from the relation $d \circ \varphi = \varphi \circ d$. □

IV.5. The Long Exact Cohomology Sequence

IV.5.1. Definition [Connecting homomorphism] Let $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ be a short exact sequence of sheaves of abelian groups on X, and let $[\sigma] \in H^q(X, \mathcal{F}'')$. Represent $[\sigma]$ by a cocycle $\sigma \in Z^q(\mathfrak{U}, \mathcal{F}'')$ for some open covering $\mathfrak{U}$. By IV.2.8, after replacing $\mathfrak{U}$ by a refinement, σ lifts to a cochain $\widetilde{\sigma} \in C^q(\mathfrak{U}, \mathcal{F})$ with $\psi_* \widetilde{\sigma} = \sigma$. Since

$$\psi_*(d\widetilde{\sigma}) = d(\psi_* \widetilde{\sigma}) = d\sigma = 0,$$

the cochain $d\widetilde{\sigma}$ takes values in $\ker \psi_* = \operatorname{im} \varphi_*$ (IV.3.2), so $d\widetilde{\sigma} = \varphi_* \tau$ for a unique cochain $\tau \in C^{q+1}(\mathfrak{U}, \mathcal{F}')$ (IV.2.5). Finally $\varphi_*(d\tau) = d(d\widetilde{\sigma}) = 0$, so $d\tau = 0$, i.e. $\tau \in Z^{q+1}(\mathfrak{U}, \mathcal{F}')$. We set

$$\delta([\sigma]) := [\tau] \in H^{q+1}(X, \mathcal{F}').$$

IV.5.2. Proposition *The map $\delta : H^q(X, \mathcal{F}'') \to H^{q+1}(X, \mathcal{F}')$ is a well-defined group homomorphism, independent of all the choices made in IV.5.1.*

Proof. To ensure that δ is well-defined, we must check that the cohomology class $[\tau]$ does not depend on (1) the choice of the lift $\widetilde{\sigma}$ for a fixed cocycle σ, and (2) the choice of the representative cocycle σ within the cohomology class $[\sigma]$. We will check these under a fixed covering $\mathfrak{U}$ (the independence under refinement follows immediately from the functoriality of the direct limit).

1. Independence of the lift $\widetilde{\sigma}$:
Suppose we choose two different lifts $\widetilde{\sigma}_1, \widetilde{\sigma}_2 \in C^q(\mathfrak{U}, \mathcal{F})$ for the same cocycle $\sigma \in Z^q(\mathfrak{U}, \mathcal{F}'')$. Then by linearity:

$$\psi_*(\widetilde{\sigma}_1 - \widetilde{\sigma}_2) = 0.$$

Hence, $\widetilde{\sigma}_1 - \widetilde{\sigma}_2 \in \ker \psi_* = \operatorname{im} \varphi_*$. Since φ_* is injective, there exists a unique cochain $\lambda \in C^q(\mathfrak{U}, \mathcal{F}')$ such that $\widetilde{\sigma}_1 - \widetilde{\sigma}_2 = \varphi_* \lambda$.

Applying the coboundary operator d to this identity, we obtain:

$$d\widetilde{\sigma}_1 - d\widetilde{\sigma}_2 = d(\varphi_* \lambda) = \varphi_*(d\lambda).$$

By definition, $d\widetilde{\sigma}_1 = \varphi_* \tau_1$ and $d\widetilde{\sigma}_2 = \varphi_* \tau_2$ for unique cocycles $\tau_1, \tau_2 \in Z^{q+1}(\mathfrak{U}, \mathcal{F}')$. Substituting these gives:

$$\varphi_* \tau_1 - \varphi_* \tau_2 = \varphi_*(d\lambda) \implies \varphi_*(\tau_1 - \tau_2 - d\lambda) = 0.$$

By the injectivity of φ_*, we conclude that $\tau_1 - \tau_2 = d\lambda$. This means that τ_1 and τ_2 differ by a coboundary, and therefore $[\tau_1] = [\tau_2]$ in $H^{q+1}(X, \mathcal{F}')$.

2. Independence of the representative σ:
Now suppose that $[\sigma] = 0 \in H^q(X, \mathcal{F}'')$, which implies that $\sigma = d\eta$ for some cochain $\eta \in C^{q-1}(\mathfrak{U}, \mathcal{F}'')$ (after passing to a refinement if necessary). We need to show that $\delta([\sigma]) = 0$.

Since ψ is surjective on stalks, after possibly refining the cover once more, η lifts to a cochain $\widetilde{\eta} \in C^{q-1}(\mathfrak{U}, \mathcal{F})$ such that $\psi_*\widetilde{\eta} = \eta$. Consider the cochain $d\widetilde{\eta} \in C^q(\mathfrak{U}, \mathcal{F})$. Its image under ψ_* is:

$$\psi_*(d\widetilde{\eta}) = d(\psi_*\widetilde{\eta}) = d\eta = \sigma.$$

Thus $d\widetilde{\eta}$ is a valid lift of σ; choose $\widetilde{\sigma} = d\widetilde{\eta}$. Computing its coboundary yields:

$$d\widetilde{\sigma} = d(d\widetilde{\eta}) = 0.$$

By definition, $d\widetilde{\sigma} = \varphi_*\tau$, so $\varphi_*\tau = 0$. By the injectivity of φ_*, we have $\tau = 0$, which immediately implies $\delta([\sigma]) = [\tau] = 0 \in H^{q+1}(X, \mathcal{F}')$. □

IV.5.3. Theorem [Long exact cohomology sequence] *Let $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ be a short exact sequence of sheaves of abelian groups on X. The sequence*

$$\begin{aligned} 0 \to H^0(X, \mathcal{F}') \to H^0(X, \mathcal{F}) \to H^0(X, \mathcal{F}'') &\xrightarrow{d^0} \\ \xrightarrow{d^0} H^1(X, \mathcal{F}') \to H^1(X, \mathcal{F}) \to H^1(X, \mathcal{F}'') &\xrightarrow{d^1} \\ \xrightarrow{d^1} H^2(X, \mathcal{F}') \to \cdots & \end{aligned}$$

is exact; more precisely, for every $q \geq 0$,

$$\cdots \to H^q(X, \mathcal{F}') \xrightarrow{\varphi_*} H^q(X, \mathcal{F}) \xrightarrow{\psi_*} H^q(X, \mathcal{F}'') \xrightarrow{d^q} H^{q+1}(X, \mathcal{F}') \xrightarrow{\varphi_*} H^{q+1}(X, \mathcal{F}) \to \cdots$$

is exact, where φ_*, ψ_* are as in IV.4.9 and d^q is the connecting homomorphism of IV.5.1.

Proof. Throughout, fix an open covering $\mathfrak{U} = (U_i)_{i \in I}$ of X; all identities are to be understood after passing to a common refinement and taking direct limits, which we suppress to lighten notation.

Exactness at $H^0(X, \mathcal{F}')$ and $H^0(X, \mathcal{F})$. By IV.4.8, $H^0(X, -)$ is identified with global sections, and the result follows directly from the left-exactness of global sections (IV.3.2).

The remaining three exactness conditions hold uniformly for all $q \geq 0$.

Exactness at $H^q(X, \mathcal{F})$: $\ker \psi_* = \operatorname{im} \varphi_*$.
($\supseteq$) Since $\psi \circ \varphi = 0$, one has $\psi_* \circ \varphi_* = 0$, so $\operatorname{im} \varphi_* \subseteq \ker \psi_*$.
($\subseteq$) Let $[\sigma] \in \ker \psi_*$, represented by $\sigma \in Z^q(\mathfrak{U}, \mathcal{F})$ with $\psi_*\sigma = d\eta$ for some $\eta \in C^{q-1}(\mathfrak{U}, \mathcal{F}'')$. Lift η to $\widetilde{\eta} \in C^{q-1}(\mathfrak{U}, \mathcal{F})$ (using IV.2.8). Then

$$\psi_*(\sigma - d\widetilde{\eta}) = d\eta - d(\psi_*\widetilde{\eta}) = 0,$$

so $\sigma - d\widetilde{\eta} \in \ker \psi_* = \operatorname{im} \varphi_*$ at the cochain level (IV.3.2). Write $\sigma - d\widetilde{\eta} = \varphi_*\tau$ for a unique $\tau \in C^q(\mathfrak{U}, \mathcal{F}')$ (IV.2.5). Then $\varphi_*(d\tau) = d\varphi_*\tau = d(\sigma - d\widetilde{\eta}) = 0$, hence $d\tau = 0$ by injectivity, and $[\sigma] = \varphi_*[\tau]$.

Exactness at $H^q(X, \mathcal{F}'')$: $\ker d^q = \operatorname{im} \psi_*$.

($\supseteq$) If $[\sigma] = \psi_*[\widetilde{\sigma}]$ with $\widetilde{\sigma} \in Z^q(\mathfrak{U}, \mathcal{F})$, choose $\widetilde{\sigma}$ itself as a lift in IV.5.1. Then $d\widetilde{\sigma} = 0$, so $\tau = 0$ and $d^q[\sigma] = 0$.
($\subseteq$) Let $[\sigma] \in \ker d^q$. In the construction of IV.5.1, lift σ to $\widetilde{\sigma} \in C^q(\mathfrak{U}, \mathcal{F})$ and obtain $\tau \in Z^{q+1}(\mathfrak{U}, \mathcal{F}')$ with $\varphi_*\tau = d\widetilde{\sigma}$ and $[\tau] = 0$. Write $\tau = d\lambda$ for some $\lambda \in C^q(\mathfrak{U}, \mathcal{F}')$. Then

$$d(\widetilde{\sigma} - \varphi_*\lambda) = d\widetilde{\sigma} - \varphi_* d\lambda = \varphi_*\tau - \varphi_*\tau = 0,$$

so $\widetilde{\sigma} - \varphi_*\lambda \in Z^q(\mathfrak{U}, \mathcal{F})$, and $\psi_*(\widetilde{\sigma} - \varphi_*\lambda) = \psi_*\widetilde{\sigma} = \sigma$. Hence $[\sigma] = \psi_*[\widetilde{\sigma} - \varphi_*\lambda]$.

Exactness at $H^q(X, \mathcal{F}')$: $\ker \varphi_* = \operatorname{im} d^{q-1}$.
($\supseteq$) If $[\tau] = d^{q-1}[\sigma]$ for some $[\sigma] \in H^{q-1}(X, \mathcal{F}'')$, then by IV.5.1 there exists a lift $\widetilde{\sigma} \in C^{q-1}(\mathfrak{U}, \mathcal{F})$ of σ such that $\varphi_*\tau = d\widetilde{\sigma}$. Hence $\varphi_*[\tau] = [d\widetilde{\sigma}] = 0$ in $H^q(X, \mathcal{F})$.
($\subseteq$) Let $[\tau] \in \ker \varphi_*$, represented by $\tau \in Z^q(\mathfrak{U}, \mathcal{F}')$ with $\varphi_*\tau = d\widetilde{\sigma}$ for some $\widetilde{\sigma} \in C^{q-1}(\mathfrak{U}, \mathcal{F})$. Set $\sigma := \psi_*\widetilde{\sigma} \in C^{q-1}(\mathfrak{U}, \mathcal{F}'')$. Then

$$d\sigma = \psi_*(d\widetilde{\sigma}) = \psi_*(\varphi_*\tau) = 0,$$

so $\sigma \in Z^{q-1}(\mathfrak{U}, \mathcal{F}'')$. In the construction of IV.5.1 applied to $[\sigma]$, the cochain $\widetilde{\sigma}$ is a valid lift of σ with $d\widetilde{\sigma} = \varphi_*\tau$, so by definition $d^{q-1}[\sigma] = [\tau]$. □

IV.6. The vanishing of higher cohomology groups

IV.6.1. Proposition [Vanishing for smooth sheaves] *Let X be a Riemann surface. Then*

$$H^q(X, \mathcal{E}^{0,0}) = 0 \qquad \text{and} \qquad H^q(X, \mathcal{E}^{0,1}) = 0$$

for all $q \geq 1$.

Proof. We prove the statement for $\mathcal{E}^{0,0}$; the proof for $\mathcal{E}^{0,1}$ is identical. Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open cover of X, and let $\{\varphi_i\}_{i \in I}$ be a smooth partition of unity subordinate to $\mathfrak{U}$.

For $\sigma \in C^q(\mathfrak{U}, \mathcal{E}^{0,0})$, $q \geq 1$, define $k\sigma \in C^{q-1}(\mathfrak{U}, \mathcal{E}^{0,0})$ by

$$(k\sigma)_{i_0 \ldots i_{q-1}} = \sum_{j \in I} \varphi_j \, \sigma_{j i_0 \ldots i_{q-1}}.$$

Each term is extended by zero outside U_j, and the sum is locally finite; hence $k\sigma$ is well-defined and smooth.

A direct computation gives

$$dk + kd = \mathrm{Id}.$$

Indeed, on $U_{i_0} \cap \cdots \cap U_{i_q}$, the terms containing $\sigma_{j i_0 \ldots \widehat{i_m} \ldots i_q}$ cancel, and the remaining term is

$$\sum_{j \in I} \varphi_j \, \sigma_{i_0 \ldots i_q} = \sigma_{i_0 \ldots i_q}.$$

Thus, if σ is a cocycle of degree $q \geq 1$, then

$$\sigma = d(k\sigma),$$

so every cocycle is a coboundary. Hence

$$H^q(\mathfrak{U}, \mathcal{E}^{0,0}) = 0$$

for all $q \geq 1$.

The same argument applies to $\mathcal{E}^{0,1}$, since multiplication of a smooth $(0,1)$-form by a smooth function is again a smooth $(0,1)$-form. Passing to the direct limit over all covers gives the result. □

IV.6.2. Proposition [Vanishing for some sheaves of meromorphic functions] *Let X be a compact Riemann surface and let D be any divisor on X. Then, for all $q \geq 2$,*

$$H^q(X, \mathcal{O}_X[D]) = 0.$$

Proof. We first prove the result for $D = 0$. Consider the Dolbeault resolution

$$0 \to \mathcal{O}_X \to \mathcal{E}^{0,0} \xrightarrow{\bar{\partial}} \mathcal{E}^{0,1} \to 0.$$

By IV.6.1, the sheaves $\mathcal{E}^{0,0}$ and $\mathcal{E}^{0,1}$ have no higher cohomology. Hence the long exact sequence in cohomology gives, for $q \geq 2$,

$$H^{q-1}(X, \mathcal{E}^{0,1}) \to H^q(X, \mathcal{O}_X) \to H^q(X, \mathcal{E}^{0,0}).$$

The two outer groups vanish, since $q - 1 \geq 1$. Therefore

$$H^q(X, \mathcal{O}_X) = 0$$

for all $q \geq 2$.

Let now A be a divisor on X and let $p \in X$. We have a short exact sequence

$$0 \to \mathcal{O}_X[A - p] \to \mathcal{O}_X[A] \to \mathbb{C}_p \to 0.$$

The associated long exact sequence gives, for $q \geq 2$,

$$H^{q-1}(X, \mathbb{C}_p) \to H^q(X, \mathcal{O}_X[A - p]) \to H^q(X, \mathcal{O}_X[A]) \to H^q(X, \mathbb{C}_p).$$

By IV.6.1, the two outer groups vanish. Thus

$$H^q(X, \mathcal{O}_X[A - p]) \cong H^q(X, \mathcal{O}_X[A])$$

for every $q \geq 2$.

Since X is compact, every divisor on X has finite support. Thus any divisor D is obtained from 0 by adding and subtracting finitely many points. Iterating the preceding isomorphism gives

$$H^q(X, \mathcal{O}_X[D]) \cong H^q(X, \mathcal{O}_X) = 0$$

for all $q \geq 2$. □

IV.6.3. Remark The intuition is that when a sheaf carries enough sections, its higher Čech cohomology groups necessarily vanish.

IV.6.4. Lemma *Let X be a space, and let $\mathcal{U} = \{U_i\}$ be an open cover of X. Then there is a collection of integer-valued functions $\{\varphi_i\}$ on X satisfying*

- *every point p in X lies in only finitely many of the support sets of the φ_i,*
- *for every point $p \in X$, $\sum_i \varphi_i(p) = 1$, and*
- $\mathrm{Supp}(\varphi_i) \subseteq U_i$ *for every i.*

Proof. Take the open covering $\{U_i\}$ and totally order the index set. Then define

$$\varphi_i(p) = \begin{cases} 1 & \text{if } p \in U_i - \bigcup_{j<i} U_j, \\ 0 & \text{otherwise.} \end{cases}$$

This collection of functions works. □

These functions are generally discontinuous of course. But if f is a section of a skyscraper sheaf $\mathcal{F}$ on an open set U, and φ is any $\mathbb{Z}$-valued function defined on U, then φf is also a section of $\mathcal{F}$ on U. Hence these are possible functions to use in partition of unity arguments involving skyscraper sheaves.

IV.6.5. Proposition [Vanishing for skyscraper sheaves] *Let $\mathcal{F}$ be a skyscraper sheaf on a space X. Then, for all $q \geq 1$,*

$$H^q(X, \mathcal{F}) = 0.$$

Proof. It suffices to show that $H^q(\mathcal{U}, \mathcal{F}) = 0$ for every open covering $\mathcal{U} = \{U_i\}_{i \in I}$ of X and every $q \geq 1$; passing to the direct limit over refinements of coverings then yields $H^q(X, \mathcal{F}) = 0$.

Fix a covering $\mathcal{U} = \{U_i\}$, and let $\{\varphi_i\}$ be an integer-valued partition of unity subordinate to $\mathcal{U}$, as constructed in IV.6.4: every point lies in only finitely many of the sets $\mathrm{Supp}(\varphi_i)$, $\sum_i \varphi_i(p) = 1$ for every $p \in X$, and $\mathrm{Supp}(\varphi_i) \subseteq U_i$.

For $q \geq 1$, define a map

$$k : C^q(\mathcal{U}, \mathcal{F}) \longrightarrow C^{q-1}(\mathcal{U}, \mathcal{F})$$

as follows. Given an q-cochain $f = (f_{i_0 \cdots i_q})$, set, for each $(q-1)$-tuple of indices $i_0, \ldots, i_{q-1}$,

$$(kf)_{i_0 \cdots i_{q-1}} := \sum_{j \in I} \varphi_j \, f_{j, i_0, \ldots, i_{q-1}}.$$

The sum over $j \in I$ is locally finite: at each point $p \in U_{i_0 \cdots i_{n-1}}$, only finitely many φ_j are supported, so only finitely many terms are nonzero near p, and the sum defines a genuine section $(kf)_{i_0 \cdots i_{n-1}} \in \mathcal{F}(U_{i_0 \cdots i_{n-1}})$. Hence $kf \in C^{n-1}(\mathcal{U}, \mathcal{F})$ is well-defined. As in the previous proof, a calculation shows that for every $q \geq 1$ and every $f \in C^q(\mathcal{U}, \mathcal{F})$,

$$d(kf) + k(df) = f.$$

If f is a cocycle, the previous homotopy relation shows that it is a coboundary and thus $H^q(\mathcal{U}, \mathcal{F}) = 0$ for every $q \geq 1$ and every covering $\mathcal{U}$. Taking the direct limit over refinements of coverings,

$$H^q(X, \mathcal{F}) = 0 \qquad \text{for all } n \geq 1.$$

□

IV.7. Riemann-Roch Theorem

IV.7.1. Definition Let $\mathcal{F}$ be a sheaf on X such that $H^q(X, \mathcal{F}) = 0$ for all q large enough. The *Euler caracteristic* of $\mathcal{F}$ is

$$\chi(\mathcal{F}) := \sum_{q=0}^{\infty} (-1)^n \dim H^q(X, \mathcal{F})$$

IV.7.2. Definition Let X be a compact Riemann surface, the dimension of $H^1(X, \mathcal{O}_X)$ is called the *analytic genus* of X and denoted g_{an}.

IV.7.3. Theorem [Riemann-Roch, first form] *Let X be a compact Riemann surface of analytic genus g_{an} and D any divisor on X. Then $H^0(X, \mathcal{O}_X[D])$ and $H^1(X, \mathcal{O}_X[D])$ are finite dimensional and the following equation holds:*

$$\dim L(D) - \dim H^1(X, \mathcal{O}_X[D]) = \deg D + 1 - g_{an}$$

Proof. First, let us establish the base case $D = 0$. By Liouville's theorem (I.2.5), any global holomorphic function on the compact surface X is constant, yielding $H^0(X, \mathcal{O}_X) \cong \mathbb{C}$, so $\dim H^0(X, \mathcal{O}_X) = 1$. By definition, the analytic genus is the dimension of the first cohomology group of the structure sheaf, $g_{an} := \dim H^1(X, \mathcal{O}_X)$, which is finite. Thus, both groups are finite-dimensional for $D = 0$, and the sheaf Euler characteristic is:

$$\chi(\mathcal{O}_X) = \dim H^0(X, \mathcal{O}_X) - \dim H^1(X, \mathcal{O}_X) = 1 - g_{an}.$$

Now, consider the short exact sequence of sheaves obtained by adding a point $p \in X$ to an arbitrary divisor $A - p$:

$$0 \to \mathcal{O}_X[A - p] \to \mathcal{O}_X[A] \xrightarrow{\mathrm{ev}_p} \mathbb{C}_p \to 0.$$

By applying the long exact sequence in cohomology (IV.5.3), and using the fact that the skyscraper sheaf $\mathbb{C}_p$ has no higher cohomology ($H^q(X, \mathbb{C}_p) = 0$ for $q \geq 1$) IV.6.5, we obtain the exact sequence:

$$\begin{aligned} 0 \to H^0(X, \mathcal{O}_X[A - p]) \to H^0(X, \mathcal{O}_X[A]) \to H^0(X, \mathbb{C}_p) \\ \xrightarrow{\delta} H^1(X, \mathcal{O}_X[A - p]) \to H^1(X, \mathcal{O}_X[A]) \to 0, \end{aligned}$$

where δ is the connecting homomorphism. Since $H^0(X, \mathbb{C}_p) \cong \mathbb{C}$ is a space of dimension 1, this sequence shows that if the cohomology groups for $A-p$ are finite-dimensional, they must also be finite-dimensional for A. Starting from 0 and adding or subtracting finitely many points, we show that the cohomology groups for A are finite-dimensional (since they are for 0). Note that for any divisor A, the higher cohomology groups $H^q(X, \mathcal{O}_X[A])$ vanish for $q \geq 2$ by IV.6.2, so sheaf Euler characteristic reduces exactly to $\chi(\mathcal{O}_X[A]) = \dim H^0(X, \mathcal{O}_X[A]) - \dim H^1(X, \mathcal{O}_X[A])$. By the rank-nullity theorem applied across the alternating sum of dimensions in this exact sequence, we obtain the key inductive relation for the Euler characteristic:

$$\begin{aligned} \chi(\mathcal{O}_X[A]) &= \chi(\mathcal{O}_X[A - p]) + \chi(\mathbb{C}_p), \\ &= \chi(\mathcal{O}_X[A - p]) + 1. \end{aligned}$$

Write $D = P - N$, where $P, N \geq 0$. Iterating our inductive relation by subtracting the points of P step-by-step starting from the sheaf $\mathcal{O}_X[D]$, and subtracting the points of N step-by-step starting from the sheaf $\mathcal{O}_X[0]$ we find:

$$\begin{aligned} \chi(\mathcal{O}_X[D]) &= \chi(\mathcal{O}_X[-N]) + \deg P, \\ \chi(\mathcal{O}_X[0]) &= \chi(\mathcal{O}_X[-N]) + \deg N. \end{aligned}$$

Subtracting these two equations and recalling that $\mathcal{O}_X[0] = \mathcal{O}_X$, we get

$$\chi(\mathcal{O}_X[D]) = \deg D + 1 - g_{an}.$$

Finally, recalling the canonical isomorphisms for global sections $H^0(X, \mathcal{O}_X[D]) \cong \mathcal{O}_X[D](X) \cong L(D)$, expanding $\chi(\mathcal{O}_X[D])$ yields the desired result:

$$\dim L(D) - \dim H^1(X, \mathcal{O}_X[D]) = \deg D + 1 - g_{an}.$$

□

IV.7.4. Corollary Let X be a compact Riemann surface of analytic genus g_{an} and D any divisor on X. Then

$$\dim L(D) \geq \deg D + 1 - g_{an}.$$

V. Serre Duality and Classical Riemann–Roch

In this section, X will always denote a compact Riemann surface, of analytic genus g_{an} and topological genus g_{top}. The last inequality established in IV.7.4 may at first seem modest at first, but it has several important consequences. We begin with a first proposition that will be useful later. We follow the approach presented in [For81].

V.0.1. Proposition *Let $p \in X$. Then there exists a nonconstant meromorphic function f on X having p as its unique pole.*

Proof. Consider the divisor $D = (g_{an} + 1) \cdot p$. By IV.7.4,

$$\dim L(D) \geq \deg D + 1 - g_{an} = 2.$$

Hence $L(D)$ contains a nonconstant meromorphic function f. Since $f \in L(D)$, we have $\operatorname{div}(f) + D \geq 0$, so every pole of f is contained in the support of D, namely at the point p. Thus f has p as its unique pole. □

V.1. The Residue Map and Mittag-Leffler Distributions

Let Ω^1_X denote the sheaf of holomorphic 1-forms on X. Recall that the short exact sequence of sheaves

$$0 \to \Omega^1_X \to \mathcal{E}^{1,0}_X \xrightarrow{\bar\partial} \mathcal{E}^{(1,1)}_X \to 0$$

induces a canonical isomorphism

$$H^1(X, \Omega^1_X) \cong \mathcal{E}^{(1,1)}_X(X)/\bar\partial\mathcal{E}^{1,0}_X(X).$$

V.1.1. Definition The *residue linear form* $\operatorname{Res} : H^1(X, \Omega^1_X) \to \mathbb{C}$ is defined by

$$\operatorname{Res}(\xi) := \frac{1}{2\pi i} \iint_X \omega,$$

where $\omega \in \mathcal{E}^{(1,1)}_X(X)$ is any smooth $(1,1)$-form representing the cohomology class ξ. By Stokes' theorem, this definition is independent of the choice of the representative ω.

To relate this global analytic residue to local algebraic data, we introduce the notion of Mittag-Leffler distributions. Let $\mathcal{M}^{(1)}_X$ be the sheaf of meromorphic 1-forms on X, and let $\mathfrak{U} = (U_i)_{i \in I}$ be an open covering of X.

V.1.2. Definition A Čech 0-cochain $\mu = (\omega_i) \in C^0(\mathfrak{U}, \mathcal{M}^{(1)}_X)$ is called a *Mittag-Leffler distribution* if, for all $i, j \in I$, the difference $\omega_j - \omega_i$ is holomorphic on $U_i \cap U_j$. Then $d\mu$ is a 1-cocycle with values in Ω^1_X, and we denote by $[d\mu] \in H^1(X, \Omega^1_X)$ its cohomology class.

For any point $a \in X$, the local residue of μ at a is defined by

$$\operatorname{Res}_a(\mu) := \operatorname{Res}_a(\omega_i),$$

where $i \in I$ is any index such that $a \in U_i$. This is well-defined, since residues of holomorphic 1-forms vanish. Since X is compact, $\operatorname{Res}_a(\mu) \neq 0$ for only finitely many points $a \in X$, so one may define the total residue

$$\operatorname{Res}(\mu) := \sum_{a \in X} \operatorname{Res}_a(\mu).$$

V.1.3. Proposition *Let μ be a Mittag-Leffler distribution of meromorphic 1-forms on a compact Riemann surface X. Then*

$$\operatorname{Res}(\mu) = \operatorname{Res}([d\mu]).$$

Proof. Write $\mu = (\omega_i)_{i\in I}$ with respect to an open covering $\mathfrak{U} = (U_i)_{i\in I}$ of X. Since X is compact, only finitely many points of X are poles of the forms ω_i. Let $a_1, \dots, a_r$ be these points. After refining the covering, we may assume that for each $\ell = 1, \dots, r$, there is an open set U_ℓ containing a_ℓ, that the sets $U_1, \dots, U_r$ are pairwise disjoint, and that each U_ℓ is equipped with a holomorphic coordinate z_ℓ centered at a_ℓ.

Set

$$U_0 := X \setminus \{a_1, \dots, a_r\}.$$

On U_0, the forms ω_i glue to a meromorphic 1-form without poles, hence to a holomorphic 1-form ω_0, because the differences $\omega_j - \omega_i$ are holomorphic on overlaps.

Let $(\rho_0, \rho_1, \dots, \rho_r)$ be a smooth partition of unity subordinate to the covering $(U_i)_{i\le r}$. Define a global smooth $(1,0)$-form on X by

$$\widetilde{\omega} := \sum_{\ell=0}^{r} \rho_\ell\, \omega_\ell.$$

Then

$$\bar\partial\widetilde{\omega} = \sum_{\ell=0}^{r} \bar\partial\rho_\ell \wedge \omega_\ell.$$

This is a smooth $(1,1)$-form representing the cohomology class $[d\mu] \in H^1(X, \Omega^1_X)$.

Therefore,

$$\operatorname{Res}([d\mu]) = \frac{1}{2\pi i}\iint_X \bar\partial\widetilde{\omega} = \frac{1}{2\pi i}\sum_{\ell=0}^{r}\iint_X \bar\partial\rho_\ell \wedge \omega_\ell.$$

By Stokes' theorem, the term corresponding to $\ell = 0$ vanishes, and for $\ell \ge 1$ we obtain

$$\iint_X \bar\partial\rho_\ell \wedge \omega_\ell = \int_{\partial U_\ell} \omega_\ell.$$

Hence, using the local residue theorem

$$\begin{aligned}
\operatorname{Res}([d\mu]) &= \frac{1}{2\pi i}\sum_{\ell=1}^{r}\int_{\partial U_\ell} \omega_\ell, \\
&= \sum_{\ell=1}^{r} \operatorname{Res}_{a_\ell}(\omega_\ell), \\
&= \sum_{a\in X} \operatorname{Res}_a(\mu), \\
&= \operatorname{Res}(\mu).
\end{aligned}$$

□

V.2. The Sheaves $\Omega^1_X[D]$ and Serre's Duality

We now incorporate divisors into this framework. For any divisor $D \in \mathrm{Div}(X)$, we denote by $\Omega^1_X[D]$ the sheaf of meromorphic 1-forms such that for any open set $U \subset X$,

$$\Omega^1_X[D](U) = \{\omega \in \mathcal{M}^{(1)}(U) : \mathrm{ord}_x(\omega) \geq -D(x) \text{ for all } x \in U\}.$$

In particular, $\Omega^1_X[0] = \Omega^1_X$ the sheaf of holomorphic 1-forms. If $\omega_0 \in \mathcal{M}^{(1)}(X)$ is a non-trivial global meromorphic 1-form with canonical divisor $K = \mathrm{div}(\omega_0)$, then multiplication by ω_0 induces a sheaf isomorphism $\mathcal{O}_X[D+K] \cong \Omega^1_X[D]$ (see III.4.5).

The natural multiplication map of sheaves $\Omega^1_X[-D] \times \mathcal{O}_X[D] \to \Omega^1_X$ defined by $(\omega, f) \mapsto f\omega$ induces a bilinear mapping on cohomology groups:

$$H^0(X, \Omega^1_X[-D]) \times H^1(X, \mathcal{O}_X[D]) \longrightarrow H^1(X, \Omega^1_X).$$

Composing this with the residue linear form Res yields a canonical pairing.

V.2.1. Definition The *Serre duality pairing* $\langle \cdot, \cdot \rangle : H^0(X, \Omega^1_X[-D]) \times H^1(X, \mathcal{O}_X[D]) \to \mathbb{C}$ is defined by

$$\langle \omega, \xi \rangle := \mathrm{Res}(\omega\xi).$$

This bilinear pairing induces a canonical linear mapping ι_D from $H^0(X, \Omega^1_X[-D])$ into the algebraic dual of the first cohomology group:

$$\begin{array}{rccc} \iota_D : & H^0(X, \Omega^1_X[-D]) & \longrightarrow & H^1(X, \mathcal{O}_X[D])^* \\ & \omega & \longmapsto & \langle \omega, \cdot \rangle \end{array}$$

The Serre Duality theorem states that the pairing $\langle \cdot, \cdot \rangle$ establishes an isomorphism between these spaces.

V.2.2. Theorem [Serre duality] *For any divisor D on a compact Riemann surface X, the linear map*

$$\iota_D : H^0(X, \Omega^1_X[-D]) \longrightarrow H^1(X, \mathcal{O}_X[D])^*$$

is an isomorphism of finite-dimensional complex vector spaces.

Proof. ι_D is injective. Let $\omega \in H^0(X, \Omega^1_X[-D]) \setminus \{0\}$, we show that there exists a $\chi \in H^1(X, \mathcal{O}_X[D])$ such that $\langle \omega, \chi \rangle \neq 0$. Let $a \in X$ be a point such that $D(a) = 0$, and let U_0 be a coordinate neighborhood of a with a local coordinate z centered at a such that $D|_{U_0} = 0$. On U_0, we may write $\omega = f\,dz$ where $f \in \mathcal{O}_X(U_0)$. Shrinking U_0 if necessary, we may assume f has no zeros on $U_0 \setminus \{a\}$, and that it does not meet $\mathrm{Supp}(D) \setminus \{a\}$.

Set $U_1 = X \setminus \{a\}$ and define the open cover $\mathfrak{U} = (U_0, U_1)$. Consider the 0-cochain $\eta = (\eta_0, \eta_1) \in C^0(\mathfrak{U}, \mathcal{M}_X)$ given by $\eta_0 = (zf)^{-1}$ and $\eta_1 = 0$. The product $\omega\eta$ is a Mittag-Leffler distribution of meromorphic 1-forms with $\omega\eta_0 = \dfrac{dz}{z}$ and $\omega\eta_1 = 0$.

Its coboundary $d\eta = \eta_1 - \eta_0 = -(zf)^{-1}$ on $U_0 \cap U_1$ belongs to $Z^1(\mathfrak{U}, \mathcal{O}_X[D])$ since $D(a) = 0$ and f is non-zero on $U_0 \setminus \{a\}$. Let $\chi = [d\eta] \in H^1(X, \mathcal{O}_X[D])$ be its cohomology class. Since $\omega\chi = [d(\omega\eta)]$, we obtain by the previous proposition:

$$\langle \omega, \chi \rangle = \mathrm{Res}(\omega\chi) = \mathrm{Res}(\omega\eta) = \mathrm{Res}_a\left(\frac{dz}{z}\right) = 1 \neq 0.$$

This proves that ι_D is injective.

ι_D is surjective. Let $\lambda \in H^1(X, \mathcal{O}_X[D])^*$ be non-zero. We must show $\lambda \in \mathrm{Im}(\iota_D)$. Let $p \in X$ be a point and define the divisor $D_n = D - n \cdot p$ for $n \in \mathbb{N}$.

Consider the subspace $\Lambda = \{\psi\lambda \mid \psi \in H^0(X, \mathcal{O}_X[n \cdot p])\} \subset H^1(X, \mathcal{O}_X[D_n])^*$. Since multiplication by a non-zero ψ induces an injective linear map on the duals (as $\mathcal{O}_X[D_n] \hookrightarrow \mathcal{O}_X[D]$ induces a surjection on H^1), Λ is isomorphic to $H^0(X, \mathcal{O}_X[n \cdot p])$. By the Riemann–Roch inequality (IV.7.4), we have $\dim \Lambda \geq n + 1 - g_{an}$.

On the other hand, the image of ι_{D_n} satisfies

$$\dim \mathrm{Im}(\iota_{D_n}) = \dim H^0(X, \Omega^1_X[-D_n]) \geq n + k_0 - \deg D$$

for some constant integer k_0. For $n > \deg D$, $\deg D_n < 0$, which implies $H^0(X, \mathcal{O}_X[D_n]) = 0$. The Riemann–Roch theorem then gives $\dim H^1(X, \mathcal{O}_X[D_n])^* = n + g_{an} - 1 - \deg D$.

For n sufficiently large, we obtain:

$$\dim \Lambda + \dim \mathrm{Im}(\iota_{D_n}) > \dim H^1(X, \mathcal{O}_X[D_n])^*.$$

This strict dimension inequality forces the intersection $\Lambda \cap \mathrm{Im}(\iota_{D_n})$ to be non-trivial. Thus, there exist a non-zero $\psi \in H^0(X, \mathcal{O}_X[n \cdot p])$ and an $\omega \in H^0(X, \Omega^1_X[-D_n])$ such that $\psi\lambda = \iota_{D_n}(\omega)$.

Setting $\omega_0 = \psi^{-1}\omega$, the naturality of the pairing yields $\lambda = \iota_D(\omega_0)$. It easily follows from the definition of the sheaves that $\omega_0 \in H^0(X, \Omega^1_X[-D])$. Hence ι_D is surjective. □

V.2.3. Corollary For any divisor D and canonical divisor K on X, we have the equality of dimensions:

$$\dim H^1(X, \mathcal{O}_X[D]) = \dim H^0(X, \Omega^1_X[-D]) = \dim L^{(1)}(-D) = \dim L(K - D).$$

V.2.4. Theorem [Riemann-Roch : classical form] *Let D be any divisor on X, and K a canonical one. Then*

$$\dim L(D) - \dim L(K - D) = \deg D + 1 - g_{an}.$$

V.2.5. Theorem [Equality of the genera] *We have $g_{an} = g_{top}$.*

Proof. Note that there is a non trivial canonical divisor K thanks to V.0.1. Applying V.2.4 to $D = K$ we get

$$\dim L(K) - \dim L(K - K) = \deg K + 1 - g_{an}.$$

Recall that $\dim L(K) = \dim H^1(X, \mathcal{O}_X) = g_{an}$ and that $\deg K = 2g_{top} - 2$ by III.2.6. The result follows. □

VI. The Consequences of Riemann-Roch Theorem

VI.1. The Vanishing of $H^1(X, \mathcal{O}_X[D])$ for Large Degree

VI.1.1. Theorem *Let X be a compact Riemann surface of genus g, and let D be a divisor on X. If $\deg D \geq 2g-1$, then*

$$H^1(X, \mathcal{O}_X[D]) = 0.$$

Proof. Let K be any non constant canonical divisor on X. Recall from Serre's duality theorem V.2.2 that $H^1(X, \mathcal{O}_X[D]) \cong H^0(X, \Omega^1_X[-D]) \cong L(K-D)$. Using the inequality given on $\deg D$ and III.2.6 we get

$$\deg(K-D) = \deg K - \deg D \leq -1.$$

Using III.4.7, the result follows.

□

VI.1.2. Corollary For every divisor D with $\deg D \geq 2g-1$, one has

$$\dim L(D) = \deg D + 1 - g.$$

One of the consequence of the previous corollary is the vanishing for higher cohomology groups.

VI.2. The Projective Embedding

VI.2.1. Theorem *Every compact Riemann surface admits a holomorphic embedding into some projective space $\mathbb{P}^N$.*

Proof. Let X be a compact Riemann surface of genus g, and let D be a divisor of degree $d \geq 2g+1$. Set $n = d-g$; by Corollary VI.1.2, for any $p, q \in X$ (not necessarily distinct):

$$\dim L(D) = n+1, \qquad \dim L(D-p) = n, \qquad \dim L(D-p-q) = n-1. \qquad (\diamondsuit)$$

Fix a basis $(f_0, \ldots, f_n)$ of $L(D)$ and define

$$\phi_D \colon X \to \mathbb{P}^n, \quad x \mapsto [f_0(x) : \cdots : f_n(x)],$$

evaluating each $f_i(x)$ via a local trivialisation of $\mathcal{O}(D)$ at x. If some $x \in X$ were a base point (i.e. $f_i \in L(D-x)$ for all i), then $L(D) = L(D-x)$, contradicting ($\diamondsuit$). Hence ϕ_D is a well-defined holomorphic map. Since any two bases of $L(D)$ yield maps differing by a projective automorphism of $\mathbb{P}^n$, injectivity and the immersion condition may be verified for any convenient basis.

Injectivity. Let $p \neq q$. By ($\diamondsuit$) one may choose $f_0 \in L(D) \setminus L(D-p)$ and $f_1 \in L(D-p) \setminus L(D-p-q)$. Since $\operatorname{ord}_p(f_0) = -D(p) < \operatorname{ord}_p(f_j)$ for all $j \geq 1$, one has $\phi_D(p) = [1 : 0 : \cdots : 0]$. Suppose $\phi_D(q) = [1 : 0 : \cdots : 0]$; then $\operatorname{ord}_q(f_j) > \operatorname{ord}_q(f_0) \geq -D(q)$ for all $j \geq 1$, so $f_j \in L(D-q)$ for $j \geq 1$. In particular,

$$f_1 \in L(D-p) \cap L(D-q) = L(D-p-q),$$

contradicting the choice of f_1. Hence $\phi_D(p) \neq \phi_D(q)$.

Immersion. Fix $p \in X$. By ($\diamondsuit$) one may choose $f_0 \in L(D) \setminus L(D-p)$ and $f_1 \in L(D-p) \setminus L(D-2p)$. Since $\operatorname{ord}_p(f_1) = \operatorname{ord}_p(f_0) + 1$, in local coordinates z at p one obtains

$$\phi_D(z) = \big[1 : z\, g_1(z) : h_2(z) : \cdots : h_n(z)\big],$$

where $g_1, h_2, \ldots, h_n$ are holomorphic at 0 with $g_1(0) \neq 0$. The derivative of the first affine coordinate $z \mapsto z\, g_1(z)$ equals $g_1(0) \neq 0$ at $z = 0$, so ϕ_D is an immersion at p.

Since ϕ_D is a holomorphic injective immersion from the compact space X into the Hausdorff space $\mathbb{P}^n$, it is a holomorphic embedding. □

The previous result can in fact be improved in the following manner. This is because the $\mathbb{C}$-dimension of X is one.

VI.2.2. Remark Every compact Riemann surface X can be embedded holomorphically into $\mathbb{P}^3$.

VII. The GAGA Theorem for Curves

This section is devoted to Serre's GAGA theorem for curves. The starting point is Chow's theorem, which guarantees that a compact Riemann surface, once embedded in projective space, is in fact an *algebraic* curve; this is the fact that makes it meaningful to speak of an algebraic structure on X at all, and it is therefore the natural place to begin. We then construct this algebraic structure — the Zariski topology and the sheaves $\mathcal{O}_{X,\mathrm{alg}}[D]$ — and finally show that its cohomology agrees with the analytic cohomology already studied.

VII.1. Chow's Theorem

VII.1.1. Definition [Closed Analytic Subvariety] A subset $V \subset \mathbb{P}^n$ is called an *analytic subvariety* of $\mathbb{P}^n$ if it is locally cut out by holomorphic functions. That is, for every point $p \in \mathbb{P}^n$, there exists an open neighborhood $U \subset \mathbb{P}^n$ containing p and a finite collection of holomorphic functions $f_1, f_2, \ldots, f_k : U \longrightarrow \mathbb{C}$ such that:

$$V \cap U = \{z \in U \mid f_1(z) = f_2(z) = \cdots = f_k(z) = 0\}\,.$$

VII.1.2. Theorem [Remmert's Proper Mapping Theorem] *Let X and Y be complex analytic spaces, and let*

$$f : X \longrightarrow Y$$

be a proper holomorphic map. Then the image $f(X)$ is a closed analytic subset of Y. More generally, if $A \subset X$ is a closed analytic subset of X, then $f(A)$ is a closed analytic subset of Y.

VII.1.3. Remark We admit VII.1.2 without proof; see [Rem57] for the original argument, or [GR65] for a textbook treatment. Its proof relies on local analytic geometry in several complex variables — Weierstrass preparation and the local structure of holomorphic germs — which lies outside the Riemann surface techniques developed in this report, and is not itself illuminating for our purposes. By contrast, the proof of VII.1.4 below is given in full, since it is precisely the argument that shows how properness lets one pass from analytic data to the polynomial equations defining C as a projective variety — the concrete mechanism this report is building toward.

VII.1.4. Theorem [Chow, Curve Case] *Let $C \subset \mathbb{P}^N$ be a closed analytic subset of dimension 1. Then C is a projective algebraic curve: there exist homogeneous polynomials $P_1, \ldots, P_r \in \mathbb{C}[X_0, \ldots, X_N]$ such that*

$$C = \{[x_0 : \cdots : x_N] \in \mathbb{P}^N \mid P_1(x) = \cdots = P_r(x) = 0\}.$$

Proof. The proof proceeds in six steps: we construct an affine cone V over C in $\mathbb{C}^{N+1}$, establish its analyticity (the central analytic argument), use the cone structure to extract homogeneous polynomial equations for V, and finally descend back to $\mathbb{P}^N$.

Step 1: The affine cone.

Let $\sigma : \mathbb{C}^{N+1} \setminus \{0\} \to \mathbb{P}^N$ be the canonical projection and set

$$V^\times := \sigma^{-1}(C) \subset \mathbb{C}^{N+1} \setminus \{0\}, \qquad V := V^\times \cup \{0\} \subset \mathbb{C}^{N+1}.$$

Since C is compact, $V^\times$ is closed in $\mathbb{C}^{N+1}\setminus\{0\}$. By construction V is a *cone*: if $z\in V$ then $\lambda z\in V$ for every $\lambda\in\mathbb{C}$. Finally, $V^\times$ has complex dimension 2: it is the pullback of a one dimensional space where the fibers are themselves one dimensional.

Step 2: Analyticity of $V^\times$ in $\mathbb{C}^{N+1}\setminus\{0\}$.

Cover $\mathbb{P}^N$ by the standard affine charts $U_i := \{X_i \neq 0\}$, $0\le i\le N$. On each $\sigma^{-1}(U_i)$, the projection σ admits the holomorphic section $s_i : U_i \to \mathbb{C}^{N+1}\setminus\{0\}$ defined by normalizing the i-th coordinate to 1. The resulting local trivialisation

$$\sigma^{-1}(U_i) \xrightarrow{\sim} \mathbb{C}^*\times U_i, \qquad z \longmapsto \big(z_i,\ [z_0:\cdots:z_N]\big)$$

identifies $V^\times\cap\sigma^{-1}(U_i)$ with $\mathbb{C}^*\times(C\cap U_i)$. Since $C\cap U_i$ is analytic in U_i by hypothesis, $V^\times\cap\sigma^{-1}(U_i)$ is analytic in $\sigma^{-1}(U_i)$. Since the U_i cover $\mathbb{P}^N$, $V^\times$ is analytic in $\mathbb{C}^{N+1}\setminus\{0\}$.

Step 3: Analyticity of V at the origin.

This is the key step. Since C is compact, we can cover it by finitely many closed discs $\bar D_1,\dots,\bar D_k$ (closed charts of C as a Riemann surface). For each α, choose an index j_α such that $D_\alpha\subset U_{j_\alpha}$, let $\varphi_\alpha:\bar D_\alpha\to C$ be the corresponding closed chart, and set

$$\tilde\varphi_\alpha := s_{j_\alpha}\circ\varphi_\alpha : \bar D_\alpha \longrightarrow \mathbb{C}^{N+1}\setminus\{0\}.$$

This is a holomorphic map with the property that the j_α-th coordinate of $\tilde\varphi_\alpha(t)$ equals 1 for all $t\in\bar D_\alpha$. Fix $\varepsilon>0$ and define

$$\Psi_\alpha : \overline{D(0,\varepsilon)}\times\bar D_\alpha \longrightarrow \mathbb{C}^{N+1}, \qquad \Psi_\alpha(\lambda,t) := \lambda\,\tilde\varphi_\alpha(t).$$

The domain $\overline{D(0,\varepsilon)}\times\bar D_\alpha$ is compact, so Ψ_α is proper. By VII.1.2, its image

$$A_\alpha := \Psi_\alpha\Big(\overline{D(0,\varepsilon)}\times\bar D_\alpha\Big)$$

is a closed analytic subset of $\mathbb{C}^{N+1}$.

We claim that

$$V\cap B(0,\varepsilon) = \Big(\bigcup_{\alpha=1}^{k} A_\alpha\Big)\cap B(0,\varepsilon).$$

($\supseteq$) For $\lambda\neq 0$, $\Psi_\alpha(\lambda,t)=\lambda\tilde\varphi_\alpha(t)$ satisfies $\sigma(\Psi_\alpha(\lambda,t))=[\tilde\varphi_\alpha(t)]=\varphi_\alpha(t)\in C$, so $\Psi_\alpha(\lambda,t)\in V^\times\subset V$. For $\lambda=0$, $\Psi_\alpha(0,t)=0\in V$.

($\subseteq$) Let $z\in V^\times$ with $|z|<\varepsilon$. Since $D_1,\dots,D_k$ cover C, there exist α and $t\in\bar D_\alpha$ with $\sigma(z)=\varphi_\alpha(t)$. Then z and $\tilde\varphi_\alpha(t)$ are two representatives of the same projective point, so $z=\lambda\,\tilde\varphi_\alpha(t)$ for some $\lambda\in\mathbb{C}^*$. Since the j_α-th coordinate of $\tilde\varphi_\alpha(t)$ equals 1, comparing j_α-th coordinates gives $\lambda=z_{j_\alpha}$. Thus $|\lambda|=|z_{j_\alpha}|\le|z|<\varepsilon$, so $(\lambda,t)\in\overline{D(0,\varepsilon)}\times\bar D_\alpha$ and $z=\Psi_\alpha(\lambda,t)\in A_\alpha$. This covers $0\in V$ trivially.

The claim shows that V near the origin is a finite union of closed analytic sets, hence analytic there. Combined with Step 2, V is analytic in all of $\mathbb{C}^{N+1}$.

Step 4: Homogeneous equations for V near the origin.

Since V is analytic near 0, there exist an open ball B centred at 0 and holomorphic functions $f_1,\dots,f_s : B\to\mathbb{C}$ such that

$$V\cap B = \{z\in B \mid f_1(z)=\cdots=f_s(z)=0\}.$$

Decompose each f_j into homogeneous components:

$$f_j(z) = \sum_{m=0}^{\infty} f_{j,m}(z), \qquad f_{j,m} \text{ homogeneous of degree } m.$$

Fix $z \in V \cap B$. Since V is a cone, $tz \in V$ for all $t \in \mathbb{C}$, and for $|t|$ small enough $tz \in V \cap B$, so $f_j(tz) = 0$. But

$$f_j(tz) = \sum_{m=0}^{\infty} t^m f_{j,m}(z).$$

This power series in t vanishes identically, so every coefficient is zero:

$$f_{j,m}(z) = 0 \quad \text{for all } m \geq 0 \text{ and all } z \in V \cap B.$$

Step 5: Globalisation by homogeneity.

Each $f_{j,m}$ is a polynomial, hence holomorphic on all of $\mathbb{C}^{N+1}$. We know $f_{j,m}$ vanishes on $V \cap B$. Because V is a cone and $f_{j,m}$ is homogeneous, we can extend this to all of V directly.

Let $w \in V$. Since V is a cone, for $t \in \mathbb{C}^*$ with $|t|$ sufficiently small, the scaled point tw lies in the ball B. Thus $tw \in V \cap B$, meaning

$$f_{j,m}(tw) = t^m f_{j,m}(w) = 0.$$

Thus, every $f_{j,m}$ is a homogeneous polynomial vanishing identically on all of V.

They generate a homogeneous ideal $\mathfrak{a} := (f_{j,m})_{j,m} \subset \mathbb{C}[z_0, \ldots, z_N]$.

$\mathbb{C}[z_0, \ldots, z_N]$ being noetherian ring, it is finitely generated by (say) $P_1, \ldots, P_r$. We obtain

$$V \cap B = \{f_{j,m} = 0 \ \forall\, j, m\} \cap B = \{P_1 = \cdots = P_r = 0\} \cap B.$$

Step 6: Descent to the projective variety.

Define

$$Y := \{[z] \in \mathbb{P}^N \mid P_1(z) = \cdots = P_r(z) = 0\}.$$

Since each P_i vanishes on V (as an element of $\mathfrak{a}$), we immediately have $C \subseteq Y$.

For the reverse inclusion, let $[w] \in Y$ and pick a representative $w \in \mathbb{C}^{N+1} \setminus \{0\}$. Each P_i is homogeneous and vanishes at w, so for any $\lambda \in \mathbb{C}^*$:

$$P_i(\lambda w) = \lambda^{\deg P_i} P_i(w) = 0.$$

Choosing $|\lambda|$ small enough ensures $\lambda w \in B$, and since $\lambda w \in \{P_1 = \cdots = P_r = 0\} \cap B = V \cap B$, we have $\lambda w \in V^\times$ (as $\lambda w \neq 0$). But V is a cone, so $w = \lambda^{-1}(\lambda w) \in V$, hence $[w] = \sigma(w) \in C$. This gives $Y \subseteq C$, and therefore $C = Y$. □

VII.1.5. Remark The theorem remains true for any closed analytic subset in a projective space.

VII.1.6. Corollary Every compact Riemann surface is isomorphic, as a complex manifold, to a smooth projective algebraic curve.

Proof. Let X be a compact Riemann surface. By VI.2.1, there exists a holomorphic embedding $f : X \hookrightarrow \mathbb{P}^n$ for some integer n. We first need to prove that its image $Y = f(X)$ is a closed analytic subset of $\mathbb{P}^n$.

Locally, because f is an embedding, its differential is everywhere injective. By the local immersion theorem for complex manifolds, for any point $x \in X$, there exist local holomorphic coordinates around x and $f(x)$ such that f takes the standard form $z \mapsto (z, 0, \ldots, 0)$. Consequently, Y is locally defined by the vanishing of $n-1$ holomorphic functions (the coordinates in the adapted local chart), meaning Y is a locally analytic subset of $\mathbb{P}^n$.

Globally, since X is compact and f is continuous, the image Y is a compact subset of $\mathbb{P}^n$. Because the projective space $\mathbb{P}^n$ is Hausdorff, Y is topologically closed.

Therefore, Y is a closed analytic subset of $\mathbb{P}^n$ of pure dimension 1. Chow's theorem then implies that Y is a projective algebraic curve.

Finally, because f is a holomorphic embedding, X is biholomorphic to its image Y. Since X is a smooth one-dimensional complex manifold, Y has no singular points. Thus, X is isomorphic to the smooth projective algebraic curve Y. □

VII.2. The Zariski Topology

In order to study the algebraic properties of a Riemann surface (or any complex manifold in general) one has to define the Zariski topology. The difficulty here is the following: how does one define polynomials on an abstract manifold? The general recipe is to define them on affine and projective spaces, and then to transport the resulting topology onto our manifold with an embedding. However that would be too much of a hurdle to do here, so we will keep it simple and only define it only on Riemann surfaces.

VII.2.1. Definition The *Zariski topology* on a compact Riemann surface X is the *cofinite topology*: a subset $Z \subseteq X$ is closed if and only if $Z = X$ or Z is finite.

VII.2.2. Remark [Compatibility with the general definition] On $\mathbb{P}^n$, a closed Zariski set is the locus of zeros of a set of homogeneous polynomials. Under any holomorphic embedding $X \hookrightarrow \mathbb{P}^n$, VII.2.1 agrees with the subspace topology induced by the projective Zariski topology of $\mathbb{P}^n$. This shows in particular that VII.2.1 does not depend on the choice of embedding.

VII.2.3. Proposition *The usual topology on $\mathbb{C}^n$ and $\mathbb{P}^n$ is finer than the Zariski one.*

Proof. The finite sets of points are closed in the classical topology because it is Hausdorff. □

VII.2.4. Remark The Zariski topology is quite counter-intuitive at first glance (e.g. it is not Hausdorff) and one must acquire an intuition ad hoc. To make things clearer, one might say that the closed sets are very small (of Lebesgue measure 0) and conversely, that open sets are all dense in the classical topology. Any two Zariski opens intersect.

When we want to talk about a Riemann surface X with the Zariski topology we will write X_{Zar}. Any open covering of X can be reduced to a finite open covering. Any two nonempty opens intersect.

VII.3. The GAGA Theorem for Meromorphic Functions with Bounded Poles

Let X be a compact Riemann surface (which we identify with a smooth projective algebraic curve). Let D be a fixed divisor on X. We denote by X_{Zar} the topological space X endowed with the Zariski topology, and by X_{an} the space X endowed with the classical (analytic) topology.

We introduce the following two sheaves:

- $\mathcal{O}_{X,alg}[D]$, the algebraic sheaf of rational functions whose poles are bounded by D.
- $\mathcal{O}_{X,an}[D]$, the analytic sheaf of meromorphic functions whose poles are bounded by D.

These sheaves are themselves subsheaves of the sheaves $\mathcal{M}_{alg}$ and $\mathcal{M}_{an}$, of the rational functions and the meromorphic functions. Their cohomology can be easily studied with the resolution introduced in the following proof. One could define them as the direct limit (taken on D) of the $\mathcal{O}_X[D]$ sheaves, but easier computation is presented here.

VII.3.1. Proposition [Vanishing of analytic H^n for unbounded poles, $n \geq 1$] *For every $n \geq 1$,*

$$H^n(X, \mathcal{M}_{an}) = 0.$$

Proof. Fix any divisor D of degree $\geq 2g-1$ on X. Consider the short exact sequence of sheaves

$$0 \to \mathcal{O}[D] \to \mathcal{M} \xrightarrow{\alpha_D} \mathcal{T}[D] \to 0,$$

where α_D sends a meromorphic function to its principal part along the poles allowed by D, and $\mathcal{T}[D]$ is the skyscraper sheaf supported on the (finite) support of D, recording these principal parts. By IV.6.5:

$$H^n(X, \mathcal{T}[D]) = 0 \qquad \text{for all } n \geq 1.$$

The long exact cohomology sequence associated to the short exact sequence above thus reads, in degrees 0 and 1,

$$0 \to \mathcal{O}[D](X) \to \mathcal{M}(X) \xrightarrow{\alpha_D(X)} \mathcal{T}[D](X) \to \underbrace{H^1(X, \mathcal{O}[D])}_{=0} \to H^1(X, \mathcal{M}) \to H^1(X, \mathcal{T}[D]) = 0,$$

so $H^1(X, \mathcal{M}) = 0$.

For higher cohomology groups ($n \geq 2$), combining this with the $n = 1$ case above, $H^n(X, \mathcal{M}_{an}) = 0$ for all $n \geq 1$ □

VII.3.2. Proposition [Vanishing of algebraic H^n for unbounded poles, $n \geq 1$] *For every $n \geq 1$,*

$$H^n(X_{Zar}, \mathcal{M}_{alg}) = 0$$

Proof. We would like to copy the proof of VII.3.1, but that would require the vanishing of the $H^n(X, \mathcal{O}_X[D])$ for $n \geq 2$, which we prove later on using this proposition.

On X_{Zar}, any finite collection of non-empty Zariski opens has non-empty intersection. The sheaf of rational functions assigns to every non-empty open set $U \subset X_{Zar}$ the same field of fractions $\mathcal{M}_{alg}(X)$, since a rational function is determined everywhere by its restriction to any arbitrarily small Zariski open set. Thus

$$\mathcal{M}_{alg}(U) = \mathcal{M}_{alg}(X) \tag{♠}$$

for all opens $U \neq \emptyset$, and every restriction map $\rho_{U,V} : \mathcal{M}_{alg}(U) \to \mathcal{M}_{alg}(V)$ is the identity map.

Direct construction of a contracting homotopy. Fix a finite open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X by non-empty Zariski opens, and fix a basepoint index $i_0 \in I$. Define a map $h : C^p(\mathcal{U}, \mathcal{M}_{alg}) \to C^{p-1}(\mathcal{U}, \mathcal{M}_{alg})$, for $p \geq 1$, by inserting i_0:

$$(h\sigma)_{j_0 \cdots j_{p-1}} := \sigma_{i_0 j_0 \cdots j_{p-1}}.$$

This is well-defined precisely because any finite collection of non-empty Zariski opens has non-empty intersection and $\mathcal{M}_{alg}(U_{i_0 j_0 \cdots j_{p-1}}) = \mathcal{M}_{alg}(U_{j_0 \cdots j_{p-1}})$ by (♠), with no restriction map needed to pass between them.

A short calculation shows that

$$dh + hd = \mathrm{Id} \qquad \text{on } C^p(\mathcal{U}, \mathcal{M}_{alg}),\ p \geq 1.$$

Let $\sigma \in Z^p(\mathcal{U}, \mathcal{M}_{alg})$, $p \geq 1$. Then

$$\sigma = d(h\sigma) + h(d\sigma) = d(h\sigma),$$

so σ is a coboundary. Hence $H^p(\mathcal{U}, \mathcal{M}_{alg}) = 0$ for every $p \geq 1$ and every finite cover $\mathcal{U}$. Passing to the direct limit over all covers,

$$H^n(X_{Zar}, \mathcal{M}_{alg}) = 0 \qquad \text{for all } n \geq 1.$$

□

VII.3.3. Remark From this proposition, one may derive the Riemann–Roch theorem for the algebraic sheaves $\mathcal{O}_X[D]$, using precisely the same argument as in the analytic case.

Since every Zariski open set is open in the classical topology, the identity map on X induces a natural comparison morphism in Čech cohomology for all $n \geq 0$:

$$j^n : H^n(X_{Zar}, \mathcal{O}_{X,alg}[D]) \longrightarrow H^n(X_{an}, \mathcal{O}_{X,an}[D])$$

VII.3.4. Theorem [GAGA (simplified)] *The comparison morphism j^n is a group isomorphism for all $n \geq 0$.*

Proof. We prove the theorem by analyzing each degree n individually. By the embedding theorem (VII.1.6), X can be considered as a projective algebraic curve. Indeed, functions on X and on its embedding are simply conjugate to each other.

Case $n = 0$: Isomorphism of Global Sections. By IV.4.8, we have to compare $\mathcal{M}_{alg}(X)$, the set of rational functions (i.e. ratios F/G of homogeneous polynomials of the same degree) on X, and $\mathcal{M}_{an}(X)$, the set of global meromorphic functions on X. Clearly $\mathcal{M}_{alg}(X) \subset \mathcal{M}_{an}(X)$.

To show the reverse inclusion, let $h \in \mathcal{M}_{an}(X)$. We view h as a global holomorphic map to the Riemann sphere, $h : X \longrightarrow \mathbb{P}^1$, by sending poles to $[1 : 0]$. Consider its graph:

$$\Gamma_h = \left\{(p, h(p)) \in X \times \mathbb{P}^1\right\} \subset \mathbb{P}^N \times \mathbb{P}^1,$$

where $X \subset \mathbb{P}^N$ is the projective embedding. Via the Segre embedding

$$\begin{array}{rccc} \sigma : & \mathbb{P}^N \times \mathbb{P}^1 & \longrightarrow & \mathbb{P}^{2N+1} \\ & ([X_0 : \cdots : X_N], [T_0 : T_1]) & \longmapsto & [X_0T_0 : X_0T_1 : \cdots : X_NT_0 : X_NT_1] \end{array},$$

$\sigma(\Gamma_h)$ is a compact complex analytic subvariety of $\mathbb{P}^{2N+1}$. Moreover, $\sigma(\Gamma_h)$ is isomorphic to X so it is a curve. By Chow's Theorem, Γ_h is a closed algebraic subvariety of $\mathbb{P}^N \times \mathbb{P}^1$.

The projection onto the first factor, $\pi_1 : \Gamma_h \longrightarrow X$, is a bijective holomorphic map between smooth compact complex curves, hence a biholomorphism and an isomorphism of algebraic curves. Consequently, π_1 induces an isomorphism of function fields:

$$k(X) \xrightarrow{\sim} k(\Gamma_h).$$

The non-constant coordinate ratio T_0/T_1 on $\mathbb{P}^1$ restricts to a rational function on Γ_h, so under this isomorphism it corresponds to a rational function on X. Thus, there exist homogeneous polynomials $A, B \in \mathbb{C}[X_0, \ldots, X_N]$ of the same degree, with $A \not\equiv 0$ on X, such that on Γ_h:

$$\frac{T_0}{T_1} = -\frac{B(X_0, \ldots, X_N)}{A(X_0, \ldots, X_N)} \implies A(X_0, \ldots, X_N)T_0 + B(X_0, \ldots, X_N)T_1 = 0.$$

On the affine chart $T_1 \neq 0$, we recover $h = T_0/T_1 = -B/A$ as a ratio of homogeneous polynomials of equal degree on X. Therefore, $h \in \mathcal{M}_{alg}(X)$, establishing $\mathcal{M}_{alg}(X) = \mathcal{M}_{an}(X)$.

Since the pole-bounding condition given by D is purely pointwise and identical in both topologies, the spaces of global sections coincide:

$$H^0(X_{Zar}, \mathcal{O}_{X,alg}[D]) = H^0(X_{an}, \mathcal{O}_{X,an}[D]),$$

and the comparison morphism j^0 is the identity isomorphism.

Case $n = 1$: Comparison via Laurent Tail Divisors. In this case, we use the Laurent tails skyscraper sheaf, presented in [Mir95]. Consider the short exact sequence of sheaves relating functions to their truncated polar parts:

$$0 \longrightarrow \mathcal{O}_X[D] \longrightarrow \mathcal{M} \xrightarrow{\alpha_D} \mathcal{T}_X[D] \longrightarrow 0$$

where $\mathcal{M}$ represents the respective sheaf of functions ($\mathcal{M}_{alg}$ or $\mathcal{M}_{an}$) and $\mathcal{T}_X[D]$ denotes the skyscraper sheaf of Laurent tail divisors bounded by $-D$. This exact sequence holds in both the algebraic and analytic categories.

Applying the long exact sequence in Čech cohomology yields:

$$\mathcal{M}(X) \xrightarrow{\alpha_D} \mathcal{T}_X[D](X) \longrightarrow H^1(X, \mathcal{O}_X[D]) \longrightarrow H^1(X, \mathcal{M}). \qquad (\heartsuit)$$

By VII.3.1 and VII.3.2, $H^1(X_{an}, \mathcal{M}_{an}) = H^1(X_{Zar}, \mathcal{M}_{alg}) = 0$. Taking (♡) in the algebraic and analytic worlds gives two exact sequences

$$\mathcal{M}_{alg}(X) \xrightarrow{\alpha_{D,alg}} \mathcal{T}_{X,alg}[D](X) \longrightarrow H^1(X_{Zar}, \mathcal{O}_{X,alg}[D]) \longrightarrow 0,$$
$$\mathcal{M}_{an}(X) \xrightarrow{\alpha_{D,an}} \mathcal{T}_{X,an}[D](X) \longrightarrow H^1(X_{an}, \mathcal{O}_{X,an}[D]) \longrightarrow 0.$$

The last two terms of each sequence give

$$H^1(X_{Zar}, \mathcal{O}_{X,alg}[D]) \cong \operatorname{coker}(\alpha_{D,alg}) = \mathcal{T}_{X,alg}[D](X)/\alpha_{D,alg}(\mathcal{M}_{alg}(X)),$$
$$H^1(X_{an}, \mathcal{O}_{X,an}[D]) \cong \operatorname{coker}(\alpha_{D,an}) = \mathcal{T}_{X,an}[D](X)/\alpha_{D,an}(\mathcal{M}_{an}(X)).$$

The comparison morphism j^1 is induced by the natural inclusions. Since $\mathcal{M}_{alg}(X) = \mathcal{M}_{an}(X)$ and the global spaces of Laurent tail data are identical by definition, j^1 maps one identical cokernel directly onto the other. It is therefore a bijection, establishing the isomorphism for $n = 1$.

Case $n \geq 2$: Vanishing of Cohomology. For degrees $n \geq 2$, the isomorphism follows from the independent vanishing of both cohomology groups.

On the analytic side, IV.6.2 shows that $H^n(X_{an}, \mathcal{O}_{X,an}[D]) = 0$.

On the algebraic side, we look at the continuation of the long exact sequence for $n \geq 2$:

$$\cdots \longrightarrow H^{n-1}(X_{Zar}, \mathcal{T}_{X,alg}[D]) \longrightarrow H^n(X_{Zar}, \mathcal{O}_{X,alg}[D]) \longrightarrow H^n(X_{Zar}, \mathcal{M}_{alg}) \longrightarrow \dots$$

As established previously, the Laurent tail sheaf $\mathcal{T}_{X,alg}[D]$ is a skyscraper sheaf supported entirely on a finite set of points. By IV.6.5 and since $n \geq 2$, $H^{n-1}(X_{Zar}, \mathcal{T}_{X,alg}[D]) = 0$. For the other sheaf, VII.3.1 shows that $H^n(X_{Zar}, \mathcal{M}_{alg}) = 0$.

By exactness, $H^n(X_{Zar}, \mathcal{O}_{X,alg}[D])$ is bracketed between two zero groups and must vanish. Since both sides of the comparison map are zero, j^n is trivially an isomorphism for all $n \geq 2$. □

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